



LEONIDAS LIFTS

COMPLETE BUNDLE V3

Training Plan // Nutrition Plan // Peptide Protocol

Built on Real-World Application & Bloodwork Oversight

Written by Someone Running the Stack.

17+ YEARS NATURAL TRAINING // 405 BENCH PRESS // 225 x 27 REPS AT 200 BW

This is the complete system: Training, Nutrition, and Peptide Protocol — built on a foundation of discipline, bloodwork oversight, and published research. No shortcuts. No filler.

THIS IS NOT HEALTH ADVICE. THIS IS NOT MEDICAL ADVICE.

You **MUST** consult a licensed physician before beginning any protocol in this bundle.
All information is for research and educational purposes only.

MOST IMPORTANTLY: CONSULT YOUR DOCTOR.

◆ **BACKED BY 180 PEER-REVIEWED REFERENCES** ◆

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"Earn the foundation first. Then build on it."

THE SPARTAN SYSTEM

"This is not a collection of PDFs. This is a system — built over 17 years of natural training, tested with real bloodwork, and backed by 180 peer-reviewed references."

WHAT THIS BUNDLE IS

The Leonidas Lifts Complete Bundle is a three-part system covering every dimension of natural athletic performance: how you train, how you fuel, and how you recover at the cellular level. Every recommendation is grounded in published, peer-reviewed research — 180 citations across the three plans. Nothing is theoretical. Everything has been tested over 17+ years of natural training, validated by bloodwork, and refined through direct application.

THE THREE PILLARS

Part I — Training Plan (56 References): A complete Push/Pull/Legs 6-day split built on progressive overload principles. Every exercise is selected based on EMG research for maximal muscle activation. Includes recovery science, deload protocols, inter-set stretching, and a sauna/sleep addendum.

Part II — Nutrition Plan (68 References): Macro frameworks, meal timing, and a full supplement stack including organ support protocols (TUDCA, NAC). Features the Spartan FFMI Calculator (leonidaslifts.com) for personalized macro targets. Covers MegaFit Meals, BioGro IGF-1 timing, Amino Energy, and tier-based supplementation.

Part III — Peptide Protocol (56 References): The most comprehensive natural peptide guide available. Covers 12+ peptides across four tiers: Foundation (BPC-157, TB-500, GHK-Cu), Performance (MK-677, CJC-1295/Ipamorelin), Longevity (NAD+/NMN, MOTS-c, Epithalon, Thymosin Alpha-1, SS-31), and Specialized (Humanin, DSIP). Includes injection technique, reconstitution, bloodwork, cost estimates, and a vial dosing calculator.

HOW TO USE THIS BUNDLE

Start with the Training Plan — that is the foundation. No peptide, supplement, or meal plan will compensate for a poor training program. Layer in the Nutrition Plan once training is dialed in. The Peptide Protocol is the final layer — added only after the first two pillars are locked in and you have baseline bloodwork from your physician. **Build the pyramid from the base up. Never start from the top.**

CALCULATE YOUR FFMI: Visit leonidaslifts.com to use the Spartan FFMI Calculator. Know your fat-free mass index before setting macros or evaluating training progress.

• leonidaslifts.com



TRAINING PLAN

Push / Pull / Legs // 6-Day Split // Progressive Overload



LEONIDAS LIFTS



THE COMPLETE TRAINING PROGRAM

6-Day Push / Pull / Shoulders & Legs Split

Built on 18 Years of Natural Lifting

No Steroids. Just Work.

WITH RECOVERY SCIENCE ADDENDUM

◆ BACKED BY 56 PEER-REVIEWED REFERENCES ◆

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◆ PUSH DAY — CHEST & TRICEPS ◆

MONDAY & THURSDAY | PERFORMED 2x PER WEEK

EXERCISE	SETS	REP TARGET	NOTES
Bench Press (Progressive Overload)	9	80-100	3x15 light > 3x10 increasing > 1x8 struggle > 2x3-5 heavy
Cable Flys	3	30-45	Full stretch, controlled tempo
Light Dumbbell Fly Stretch	3	30-45	Deep stretch at bottom, light weight
Skullcrushers	4	100	Volume accumulation for tricep long head
Tricep Pulldown Trifecta (5 Cable Variations)	10	150-200	Rope / V-Bar / Straight / Reverse / Single — 2 sets each

RECOVERY: 15 MIN SAUNA | Hydrate with electrolytes | Target 80-100 C

◆ *Progressive overload on bench is the cornerstone. Track your weights. Beat the logbook every session.*

RESEARCH: WHY THESE EXERCISES

Bench Press: The flat bench press maximizes sternal and clavicular pectoralis major activation, making it the gold standard for chest development. EMG analysis shows flat bench produces the highest overall pec recruitment compared to incline variations (Rodriguez-Ridao et al., 2020). Non-uniform fiber recruitment means both flat and incline angles are valuable, but flat bench remains the primary mass builder (Cabral et al., 2021). The 9-set pyramid structure ensures progressive overload across rep ranges — light sets prime the nervous system, moderate sets accumulate volume, and heavy sets drive maximal motor unit recruitment.

Cable Flys: Unlike free-weight flys where tension drops at the top, cable flys maintain constant mechanical tension throughout the full arc of motion. Zeidler et al. (2022) demonstrated that cable-based chest exercises preserve greater shoulder joint moment across both the stretched and contracted positions. This constant tension profile makes cables superior for sustained time-under-tension and metabolic stress — two key hypertrophy drivers.

Dumbbell Fly Stretch: Stretch-mediated hypertrophy is one of the most significant findings in recent exercise science. A meta-analysis of 30 RCTs found that training at long muscle lengths produced significantly greater hypertrophy with effects greatest at long muscle lengths ($d = 0.30$, $p < 0.001$) (Warneke et al., 2024). Inter-set passive stretching may also sensitize stretch-activated signaling pathways, offering a time-efficient hypertrophic stimulus (Schoenfeld, De Souza & Wackerhage, 2022). Light dumbbell flys with a deep stretch at the bottom capitalize on this mechanism, providing a powerful hypertrophic stimulus even at low loads.

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Skullcrushers: The triceps long head crosses both the elbow and shoulder joints, meaning it is maximally stretched in overhead or inclined positions. Maeo et al. (2022) conducted a 12-week RCT showing overhead elbow extension produced +16% triceps hypertrophy vs. +10% for neutral-position extension — a clinically significant difference attributable to greater stretch on the long head during the overhead position. Skullcrushers replicate this overhead stretch mechanic by placing the elbows above the head at the bottom of each rep, putting the long head under maximal tension at its most lengthened position. This directly applies the stretch-mediated hypertrophy principle for superior triceps growth. The 4 sets of 100 total reps provide high-volume accumulation that prioritizes the long head — the largest of the three triceps heads and the primary driver of upper arm size when viewed from the side.

Tricep Pulldown Trifecta: Different elbow joint angles and grip orientations significantly alter the EMG activity-to-force ratio among the three triceps heads (Akima et al., 2021). By rotating through five cable variations — rope, V-bar, straight bar, reverse grip, and single-arm — the trifecta ensures comprehensive triceps stimulation across all heads. Barbell overhead press data confirms pressing movements produce high triceps excitation across all conditions (Ce, Coratella et al., 2022), supporting multi-angle programming for maximal triceps development. The 10-set total across five grip variations accumulates 150-200 reps of direct triceps work, providing enormous metabolic stress and cellular swelling that complement the mechanical tension from skullcrushers.

◆ PULL DAY — BACK, REAR DELTS & BICEPS ◆

TUESDAY & FRIDAY | PERFORMED 2x PER WEEK

◆ Sometimes going 20% lighter and pause-repping can make you stronger faster. Ego is the enemy of gains.

EXERCISE	SETS	REP TARGET	NOTES
Lat Pulldown Trifecta (Wide / Neutral / Close)	3 (x3 grips)	54	18 reps per grip variation
Cable Rows	3	30	Full contraction, 1s squeeze
Cable Face Pulls	2	100	High-rep posterior delt / rotator cuff
Rear Delt Machine	4	60	15 reps per set, controlled
Dumbbell Rows	3	30	10 per set, heavy and strict
Bicep Curl Trifecta	3	135	Hammers x15 > Oscillating x15 > Side Hammers x15, 30s rest, x3
Bicep Curl Machine Bi-Burn	3	60	20 reps per set, drop set on last
Cable Curls	3	150	50 reps per set — high-volume finisher

RECOVERY: 15 MIN SAUNA | Hydrate with electrolytes | Target 80-100 C

RESEARCH: WHY THESE EXERCISES

Lat Pulldown Trifecta: Grip width modulates secondary mover activation without significantly altering latissimus dorsi recruitment. Andersen et al. (2014) found shoulder-width grip produced the greatest lat activation, while Lusk et al. (2010) showed supinated grip increased biceps brachii involvement. The trifecta approach (wide, neutral, close) ensures complete lat fiber coverage and varied bicep/forearm stimulation across grips (Signorile, Zink & Szwed, 2002). Recent EMG analysis confirms posterior deltoid and biceps activation varies substantially across grip orientations (Buonsenso et al., 2025), validating the multi-grip strategy.

Cable Rows: Seated cable rows with full scapular retraction produce significantly greater middle trapezius and rhomboid activation compared to pulldown variations (Lehman et al., 2004). The horizontal pull pattern is essential for mid-back thickness and scapular health, complementing the vertical pull of lat pulldowns. Full contraction with a 1-second squeeze ensures peak scapular retractor engagement.

Face Pulls: External rotation at 90 degrees abduction maximizes posterior deltoid and infraspinatus activation (Hawkes et al., 2015). High-rep face pulls serve dual purposes: posterior deltoid hypertrophy and rotator cuff prehabilitation. The 2 sets of 100 total reps accumulate enormous time-under-tension for the external rotators, building the structural integrity needed to support heavy pressing 4x/week. This is a non-negotiable shoulder health exercise that prevents the anterior-dominant imbalances common in pressing-heavy programs.

Rear Delt Machine: Internal rotation of the humerus during lateral-raise-type movements dramatically increases posterior deltoid EMG with effect sizes of 10.79-21.34 compared to other variations (Coratella et al., 2020). Pronated grip produces greater rear delt activation than hammer grip during the eccentric phase (Demirtas et al., 2023). Machine isolation provides consistent resistance curves for targeted rear delt development.

Dumbbell Rows: Unilateral dumbbell rows allow for greater range of motion and independent limb training, addressing bilateral strength asymmetries that bilateral movements mask. The heavy, strict execution (10 reps per set) prioritizes mechanical tension on the latissimus dorsi and lower trapezius, complementing the higher-rep cable work earlier in the session. The unilateral loading also engages the obliques and quadratus lumborum as stabilizers, providing functional core work without dedicated ab training. Keep strict form — eliminating momentum ensures the target muscles bear the full load through the entire range of motion.

Bicep Curl Trifecta & Variations: Supinated grip produces 6-9% greater biceps brachii RMS activation vs. pronated grip (Coratella et al., 2023). EZ bar curls show higher combined biceps and brachioradialis activation in both concentric and eccentric phases (Paoli et al., 2018). Training at long muscle lengths (incline curls) produces greater distal biceps hypertrophy than short-length training like preacher curls (Zubillaga et al., 2023). The multi-grip trifecta cycles through hammer curls, oscillating curls, and side hammer curls to diversify both mechanical and neural stimuli for complete arm development. The high-volume cable curl finisher (50 reps per set) maximizes metabolic stress and cellular swelling, driving sarcoplasmic hypertrophy that complements the myofibrillar gains from heavier curl work. The Bicep Curl Machine bi-burn with drop sets on the last set pushes the biceps past concentric failure, recruiting additional motor units during the eccentric phase for maximal fiber damage and adaptation stimulus.

◆ SHOULDERS, TRAPS & LEGS ◆

WEDNESDAY & SATURDAY | PERFORMED 2x PER WEEK

◆ SHOULDERS & TRAPS

EXERCISE	SETS	REP TARGET	NOTES
Shoulder Burner Heavy	3	10	Heavy lateral raises — strict form
Shoulder Burner Volume	5-6	100-120	Light laterals, high rep burn
Shrugs (3s Pause at Top)	3	10	3-second hold at peak contraction
Shoulder Press (Pyramid)	5	65 total	20 / 15 / 12 / 10 / 8 reps — increasing weight

◆ LEGS

EXERCISE	SETS	REP TARGET	NOTES
Leg Extensions	3	45	15 per set, full contraction squeeze
Leg Press	4	40	10 per set, controlled depth
Hamstring Curls	3	45	15 per set, slow eccentric
Calf Raises	4	100	25 per set, full ROM + pause at stretch

RECOVERY: 15 MIN SAUNA | Hydrate with electrolytes | Target 80-100 C

WEEKLY STRUCTURE

MON & THU = Push | TUE & FRI = Pull | WED & SAT = Shoulders/Legs

SUNDAY = REST & RECOVER

RESEARCH: WHY THESE EXERCISES

Lateral Raises (Shoulder Burner): Neutral humerus rotation during lateral raise maximizes medial deltoid activation at 30.3% MVIC, outperforming frontal raise and all rotation variants (Coratella et al., 2020). Lateral raises produce significantly higher medial deltoid activation than bench press or shoulder press (Campos et al., 2020). Both dumbbell and cable lateral raises yield similar hypertrophy (3.3-4.6% thickness increase) over 8 weeks (Larsen et al., 2025), confirming either modality is effective. The heavy/volume split maximizes both mechanical tension and metabolic stress.

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Shrugs (3s Pause): Shrugs produce significantly greater upper trapezius EMG activation compared to all tested alternatives (Kim et al., 2016). High-intensity shrug loading results in sustained upper trapezius activation (Sjogaard et al., 2020). The 3-second isometric hold at peak contraction extends time-under-tension for the upper traps beyond what standard concentric/eccentric reps provide, maximizing mechanical stimulus per rep.

Shoulder Press (Pyramid): Front barbell overhead press maximizes anterior deltoid and pectoralis major excitation, while back press enhances medial and posterior deltoid activation. Barbell press uniformly exceeds machine press in overall muscle excitation (Ce, Coratella et al., 2022). Shoulder press also produces significantly higher anterior and medial deltoid activation than bench press and dumbbell fly (Campos et al., 2020). The pyramid rep scheme (20 down to 8) combines endurance, hypertrophy, and strength stimuli in a single exercise.

Leg Extensions: Systematic review confirms leg extension movements produce maximum vastus medialis (VMO) isolation, with foot position and ROM adjustments modulating quad-head-specific activation (Martin-Fuentes, Oliva-Lozano & Muyor, 2020). Full contraction squeeze at the top ensures peak quadriceps activation at the shortened position.

Leg Press: Leg press produces comparable quadriceps activation to the squat but with significantly lower erector spinae and gluteus maximus recruitment (Saeterbakken, Stien & Andersen, 2021). Foot placement modulates quad vs. glute/hamstring involvement — high narrow stance maximizes vastus lateralis activation while low stance increases gluteus maximus contribution (Martin-Fuentes et al., 2020). A quad-dominant compound without axial compression risk.

Hamstring Curls: A 10-week training study confirmed leg curls significantly increase biceps femoris long head muscle thickness and isometric torque regardless of ankle position (Cadeo et al., 2023). Nordic hamstring curl research further supports eccentric hamstring training for injury prevention and fascicle length gains across 39 studies reviewed (Sharma, Desai & Devare, 2024). Slow eccentrics amplify the hypertrophic stimulus.

Calf Raises: Standing calf raise produced 5.6% total triceps surae hypertrophy vs. 2.1% for seated over 12 weeks, driven by greater gastrocnemius activation at long muscle lengths (Kanehisa et al., 2023). Critically, partial ROM performed in the dorsiflexed (stretched) position produced 15.2% medial gastrocnemius hypertrophy vs. only 6.7% for full ROM and 3.4% for shortened position (Kassiano et al., 2023). Full ROM with emphasis on the bottom stretch is optimal.

◆ TRAINING METHODOLOGY — THE SCIENCE ◆

6-DAY PPL FREQUENCY

The Leonidas Lifts program trains each muscle group twice per week using a Push/Pull/Shoulders & Legs rotation across six training days. This frequency is directly supported by Schoenfeld, Ogborn & Krieger's (2016) meta-analysis of 22 studies, which found training each muscle group 2x/week produced significantly greater hypertrophy than 1x/week (effect size 0.37 higher), with a trend for even greater benefit at 3x/week when volume is matched. A subsequent meta-analysis (Schoenfeld, Grgic & Krieger, 2019) clarified that higher frequency primarily benefits hypertrophy by facilitating greater total weekly volume accumulation — which is the true driver of muscle growth. The 6-day structure distributes volume optimally while providing 48-72 hours between sessions for the same muscle group, matching the muscle protein synthesis recovery window.

WEEKLY VOLUME: SETS PER MUSCLE GROUP

This program delivers 18-29 direct sets per muscle group per week — well within the evidence-based optimum. Schoenfeld, Ogborn & Krieger (2017) established a dose-response relationship where 10+ sets per muscle per week produced significantly greater hypertrophy, with a continuing benefit trend up to ~20 sets/week. The Bayesian network meta-analysis by Currier et al. (2023) ranked higher-volume multiset training performed twice weekly as the highest-probability protocol for hypertrophy, with 3+ sets per exercise consistently associated with superior gains. Push day delivers ~29 sets for chest and ~14 for triceps; Pull day delivers ~24 for back and ~12 for biceps; Shoulders & Legs day delivers ~16 for shoulders and ~14 for legs — all exceeding the 10-set threshold.

PROGRESSIVE OVERLOAD

Progressive overload is the single most important training variable. Plotkin et al. (2022) conducted an 8-week RCT comparing load progression vs. repetition progression and found both produced equivalent hypertrophy and strength gains when effort was equated. This confirms that progressive overload — not specifically progressive load — is the critical variable. In this program, the bench press pyramid provides load-based progression, while high-rep exercises (cable curls, face pulls) use rep-based progression. Both pathways drive adaptation when effort is maximized.

Track every session. Beat the logbook.

REP RANGES: THE HYPERTROPHY CONTINUUM

This program intentionally spans rep ranges from 3-5 (heavy bench sets) to 50+ (cable curls). Schoenfeld et al. (2021) re-examined the repetition continuum and confirmed that low (1-5), moderate (6-12), and high (15-30+) rep ranges all produce equivalent hypertrophy when sets are taken to or near muscular failure. A volume-matched meta-analysis (Carvalho et al., 2022) reinforced that hypertrophic adaptations are equivalent across 30-90% 1RM loads. The variety of rep ranges maximizes both mechanical tension (heavy sets) and metabolic stress (high-rep sets) for comprehensive muscle development. Heavy sets additionally drive myofibrillar protein synthesis and strength gains that support future progressive overload capacity.

REST DAY PLACEMENT & RECOVERY WINDOWS

Sunday serves as the sole full rest day, strategically placed after the second Shoulders & Legs session (Saturday). Muscle protein synthesis (MPS) peaks 24-48 hours post-exercise and returns to baseline by 48-72 hours. The PPL rotation ensures each muscle group is re-stimulated within this MPS window: Push on Monday/Thursday, Pull on Tuesday/Friday, and Shoulders & Legs on Wednesday/Saturday. This maximizes the time spent in an elevated MPS state across the week. The Sunday rest day allows systemic recovery — nervous system restoration, glycogen replenishment, hormonal rebalancing, and psychological recovery — before the next training cycle begins. For athletes in a caloric deficit, this rest day is especially critical as recovery capacity is compromised.

INTENSITY TECHNIQUES: WHEN TO USE THEM

This program incorporates several intensity techniques beyond straight sets. **Drop sets** (Bicep Curl Machine) extend the set beyond initial failure by reducing load, accumulating additional metabolic stress. **Trifecta supersets** (Lat Pulldown, Bicep Curl, Tricep Pulldown) cycle through multiple grip positions with minimal rest, maximizing time-under-tension across different joint angles. **Isometric holds** (Shrug 3s pause) increase time-under-tension at peak contraction. **High-rep finishers** (Cable Curls 50 reps) drive metabolic stress and cellular swelling. These techniques should be applied as programmed — adding extra intensity techniques to already high-volume sessions risks exceeding recovery capacity, particularly during caloric deficits.

◆ RECOVERY SCIENCE ◆

RESEARCH-BACKED PROTOCOLS FOR NATURAL ATHLETES

1. SAUNA PROTOCOL

Post-workout sauna at 80-100 C activates heat shock proteins (HSP70/HSP90) that prevent muscle protein degradation and assist satellite cell repair. These molecular chaperones stabilize damaged proteins and facilitate proper refolding — essential for recovery from the high-volume training in this program.

A single 15-minute session drives HSP70/HSP72 elevation; a series of 10 Finnish sauna baths (3 x 15 min each) modulates the immune response and HSP70 family levels in both trained and untrained men (Czerwinska-Ledwig et al., 2023). The landmark Finnish study documented significant GH increases with repeated sauna exposure at 80 C — though the peak response requires longer sessions; a 15-minute post-workout protocol yields a meaningful acute GH response that stacks with exercise-induced GH release (Leppaluoto et al., 1986).

Infrared sauna reduces DOMS and preserves explosive neuromuscular performance vs. passive rest. A single 20-minute infrared session at 43 C after resistance exercise improved recovery metrics in male athletes (Ahokas et al., 2022). **In a caloric deficit:** consume 500-750 mL electrolyte fluid before/during the session and rehydrate to body weight parity. Sweat losses of 0.5-1.0 kg per session deplete sodium, potassium, and magnesium — supplement accordingly. Target 80-100 C for Finnish sauna or 43-55 C for infrared.

2. SLEEP & GH RELEASE

Approximately 70-75% of total daily GH is secreted during nocturnal sleep, with the first slow-wave sleep (SWS) episode accounting for 50-60% of the nocturnal total (Takahashi, Kipnis & Daughaday, 1968). Growth hormone is the primary anabolic hormone released during sleep — it drives muscle protein synthesis, fat mobilization, and connective tissue repair. Disrupting the first SWS cycle has a disproportionate impact on GH output.

A single night of sleep deprivation drops muscle protein synthesis by ~18%, increases cortisol ~21%, and decreases testosterone ~22% (Lamon et al., 2020). On a 6-day high-volume split, **7.5-8 hours minimum** ensures 4-5 full sleep cycles and sufficient SWS for peak GH release. Cutting to 6 hours reduces total SWS by 20-30%, directly compromising recovery capacity and hormonal optimization.

Magnesium supplementation (300-400 mg, 30-60 min before bed) improves sleep efficiency, reduces cortisol, and increases melatonin. Magnesium acts as an NMDA antagonist and GABA-A agonist, promoting neurological relaxation. An 8-week RCT in 46 subjects showed 500 mg/day magnesium significantly improved ISI score, sleep efficiency, and early morning awakening (Abbasi et al., 2012). Pair with a consistent sleep schedule — irregular bedtimes fragment SWS architecture even when total hours are adequate.

3. DELOAD PROTOCOLS

On a 6-day split, **deload every 4-5 weeks** during a caloric deficit (6-8 weeks in a surplus). An international Delphi consensus panel reached 100% agreement that deloads must be included in each mesocycle, with timing responsive to athlete fatigue markers (Strafford et al., 2023). The consensus defines deloading as "a period of reduced training stress designed to mitigate physiological and psychological fatigue, promote recovery, and enhance preparedness for subsequent training." Volume reduction of 30-50% with maintained or reduced intensity was the consensus recommendation.

Active recovery (maintain frequency, reduce volume 50-60% and intensity 30-40%) is marginally superior to complete rest — both preserve strength, but active deloads show slightly higher molecular remodeling markers including MHC-IIa mRNA expression (Vann et al., 2021). During deficit deloads, consider raising calories to maintenance temporarily to restore anabolic signaling without compromising fat loss trajectory (Pinero & Schoenfeld, 2024). A well-timed deload re-sensitizes the muscle to the training stimulus, allowing the next mesocycle to drive new adaptation.

4. CONNECTIVE TISSUE

Tendons adapt far slower than muscle (collagen half-life: 60-100 years vs. muscle protein turnover in days). Joint warning signs are **leading indicators** of overtraining — pain, clicking, or stiffness that persists beyond warmup signals accumulated connective tissue strain. On a 6-day high-volume split, proactive joint care is non-negotiable.

UC-II Collagen 40 mg/day works via oral tolerance — it activates regulatory T-cells that suppress joint inflammation. An RCT in healthy volunteers showed the UC-II group could exercise 2x longer before knee pain onset (2.8 vs. 1.4 min, $p < 0.05$) and showed significantly improved knee extension at 120 days (Lugo et al., 2013). **Aquamin** (2400 mg/day seaweed-derived minerals) significantly improved knee pain, stiffness, and walking distance (+101 feet, +7% vs baseline) in OA patients —

superior to glucosamine sulfate and placebo (Frestedt et al., 2008).

BPC-157 & TB-500: Robust preclinical data for tendon healing and angiogenesis, but **no published human RCTs** exist as of 2026. Both are WADA-banned and constitute unregulated self-experimentation. An injectable peptide review confirms BPC-157 shows potential for tendon/muscle/bone repair in animal models, while TB-500 (thymosin beta-4 fragment) promotes angiogenesis, but current evidence is insufficient to support clinical recommendation (Mayfield et al., 2026). The evidence-based position: use proven interventions (UC-II, Aquamin, vitamin C, load management) while monitoring for eventual human trial data.



“Seventeen years under the bar before I ever picked up a syringe. Earn the foundation first. Then build on it.”

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NO STEROIDS. JUST WORK.

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NUTRITION PLAN

Macros // Meal Timing // Supplements // Organ Support



LEONIDAS LIFTS

NUTRITION PLAN

13-Week Phased Cut Protocol

Progressive Calorie Reduction // High Protein // Performance Fueling

"Discipline Is the Diet."



◆ BACKED BY 68 PEER-REVIEWED REFERENCES ◆

PERSONALIZE YOUR MACROS

Use the Spartan FFMI Calculator at leonidaslifts.com — enter your height, weight, body fat %, and supplement/peptide status to get your exact macros tailored to your physique.

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PHASE OVERVIEW

13-WEEK CALORIC PERIODIZATION STRATEGY

PHASE 1

DEFICIT ENTRY

2,300

KCAL / DAY

P: 210g C: 220g F: 65g

37/38/25%

Weeks 1-8

PHASE 2

DEEP CUT

2,000

KCAL / DAY

P: 210g C: 165g F: 55g

42/33/25%

Weeks 9-12

PHASE 3

FINISHING STRIKE

1,800

KCAL / DAY

P: 200g C: 140g F: 50g

44/31/25%

Week 13

WEEK-BY-WEEK CALORIE SCHEDULE

WK 1	WK 2	WK 3	WK 4	WK 5	WK 6	WK 7
2,300	2,300	2,300	2,300	2,300	2,300	2,300
WK 8	WK 9	WK 10	WK 11	WK 12	WK 13	
2,300	2,000	2,000	2,000	2,000	1,800	

RESEARCH: WHY PROGRESSIVE CUTS WORK

Progressive caloric reduction is supported by the MATADOR protocol and adaptive thermogenesis research. Constant deficits lose effectiveness as the body adapts — phased cuts stay ahead of metabolic slowdown. Helms et al. (2014, Int J Sport Nutr Exerc Metab) established the foundational rule: target 0.5-1.0% body weight loss per week to maximize muscle retention. Rosenbaum & Leibel (2010, Int J Obes) identified adaptive thermogenesis as a coordinated set of metabolic, neuroendocrine, and autonomic responses defending body fat stores. The MATADOR study (Byrne et al., 2018, Int J Obes) demonstrated that alternating deficit/maintenance blocks produced less adaptation and greater fat loss than continuous restriction. Jorgensen et al. (2018, Br J Nutr) showed RMR suppression begins at as little as 5% body weight loss, reinforcing the need for phased approaches. For a 200 lb lifter: 0.5%/wk = ~1 lb/wk (conservative), 1.0%/wk = ~2 lbs/wk (upper bound).

PHASE 1 — DEFICIT ENTRY

WEEKS 1-8 | 2,300 KCAL/DAY

PROTEIN 210g (37%)

CARBS 220g (38%)

FAT 65g (25%)

DAILY MEAL TEMPLATE

~2,260 cal actual

1 MEAL 1 — BREAKFAST

1.5 cups egg whites + 1 whole egg, 1/2 cup oatmeal (dry), 1 banana

P: 35g C: 55g F: 8g | ~430 cal

2 MEAL 2 — MID-MORNING

1 scoop whey protein, 1 apple, 10 almonds

P: 30g C: 30g F: 8g | ~310 cal

3 MEAL 3 — LUNCH

8oz chicken breast, 1 cup white rice (cooked), broccoli

P: 50g C: 45g F: 5g | ~425 cal

4 MEAL 4 — PRE-WORKOUT

6oz ground turkey (93% lean), 1/2 cup sweet potato, spinach

P: 35g C: 30g F: 10g | ~350 cal

5 MEAL 5 — POST-WORKOUT

1.5 scoops whey protein, 1 banana, 8oz Oikos Triple Zero yogurt

P: 45g C: 45g F: 4g | ~395 cal

6 MEAL 6 — DINNER

6oz tilapia or flank steak, 1/2 cup rice, vegetables, hot sauce

P: 35g C: 25g F: 12g | ~350 cal

WHY THESE FOODS

- Egg whites: PDCAAS = 1.0 — the highest possible protein quality score (FAO, 2013). Complete essential amino acid profile with high bioavailability.
- Chicken breast: ~31g protein per 100g with minimal fat (~3.5g). The lean protein gold standard for body recomposition (USDA FoodData Central).
- White rice over brown (post-workout): Higher GI (~72 vs. ~50-68 for brown) enables rapid glycogen replenishment post-training (Atkinson et al., 2008).
- Oikos Triple Zero: Casein-dominant protein provides sustained amino acid release. Contains probiotics supporting gut health during caloric restriction.

RESEARCH: LEUCINE THRESHOLD & PROTEIN DISTRIBUTION

The leucine threshold for maximal MPS is 3-4g per meal, equivalent to ~25-35g of quality protein. Mamerow et al. (2014, J Nutr) demonstrated that evenly distributing protein across meals stimulated ~34% more 24-hour MPS than a skewed pattern with the same total intake. This 6-meal template ensures each meal hits the leucine threshold, maximizing the anabolic response during caloric restriction. Stark et al. (2012, JISSN) confirmed protein timing across 4-6 daily feedings optimizes nitrogen balance and MPS during energy deficit.

PHASE 1 — DEFICIT ENTRY (CONTINUED)

FOOD SCIENCE & NUTRIENT TIMING

FOOD SCIENCE: EVERY INGREDIENT EARNS ITS PLACE

Whole Eggs (Meal 1)

Whole eggs produce ~40% greater MPS than equivalent protein from egg whites alone (van Vliet et al., 2017, Am J Clin Nutr). The yolk lipids enhance intramuscular amino acid delivery. One whole egg + egg whites gives the best of both worlds: high volume protein with enhanced MPS signaling.

Oatmeal (Meal 1)

Beta-glucan fiber slows carbohydrate absorption, providing sustained energy through the morning (Wolever et al., 2010, J Nutr). Supports muscle glycogen stores for afternoon training sessions.

Bananas (Meals 1 & 5)

Comparable to sports drinks for exercise fueling (Nieman et al., 2012, PLoS One). ~422mg potassium per medium banana supports electrolyte balance during deficit-induced fluid shifts.

Whey Protein (Meals 2, 5)

Rapidly absorbed, high in leucine (~2.5g per scoop), and strongly stimulates MPS (Tang et al., 2009, J Appl Physiol). Post-workout timing capitalizes on the enhanced anabolic sensitivity window.

Broccoli (Meals 3, 6)

Contains DIM (diindolylmethane), which modulates estrogen metabolism by increasing the favorable 2-OHE1/16a-OHE1 ratio (Dalesandri et al., 2004, Nutr Cancer). Valuable during caloric restriction when hormonal optimization matters most.

Spinach (Meal 4)

Dietary nitrates convert to nitric oxide, improving exercise performance and oxygen efficiency (Larsen et al., 2007, Acta Physiol). Pre-workout placement maximizes training benefit.

Sweet Potato (Meal 4)

Moderate GI (~64) provides sustained energy for training. Rich in beta-carotene (8,509 mcg/100g) supporting immune function during caloric restriction (Atkinson et al., 2008, Diabetes Care).

Flank Steak / Beef (Meal 6)

Natural source of creatine (~4-5g/kg raw), heme iron, zinc, and B12 (Harris et al., 1997, Proc Nutr Soc). Heme iron absorption is 15-35% vs. 2-20% for non-heme.

Hot Sauce (Meal 6)

Capsaicin has a dose-dependent thermogenic effect (Whiting et al., 2012, Appetite). While Frank's RedHot contains modest capsaicin (~0.5mg/tbsp), the effect is additive over time and zero-calorie flavor weapons increase dietary compliance.

NUTRIENT TIMING RATIONALE

Carbohydrates are front-loaded (Meals 1, 3) and placed around training (Meals 4, 5) to support glycogen availability when it matters most. Fats are distributed with breakfast and dinner to enhance fat-soluble vitamin absorption from the supplement stack. The 6-meal structure maintains a protein feeding every 2.5-3 hours, keeping each bolus within the 25-45g range that maximizes per-meal MPS (Areta et al., 2013, J Physiol). Casein from Oikos Triple Zero in Meal 5 provides sustained amino acid release through the post-workout window.

PHASE 2 — DEEP CUT

WEEKS 9-12 | 2,000 KCAL/DAY

PROTEIN 210g (42%)

CARBS 165g (33%)

FAT 55g (25%)

KEY CHANGES FROM PHASE 1

- Carbs drop from 220g to 165g (-55g) — primarily from rice portions and oatmeal
- Fat reduced from 65g to 55g — trim cooking oils, reduce nut portions
- Protein maintained at 210g — muscle preservation is non-negotiable
- Remove 1 banana, reduce rice portions by ~1/3 across meals
- Add 2-day carbohydrate refeeds every 10-14 days to restore leptin and thyroid function

ADJUSTED MEAL TEMPLATE

~1,985 cal actual

1 MEAL 1 — BREAKFAST

1.5 cups egg whites + 1 whole egg, 1/3 cup oatmeal (dry)

P: 33g C: 30g F: 7g | ~315 cal

2 MEAL 2 — MID-MORNING

1 scoop whey protein, 1 rice cake + 1 tbsp almond butter

P: 28g C: 18g F: 10g | ~275 cal

3 MEAL 3 — LUNCH

8oz chicken breast, 2/3 cup white rice (cooked), broccoli/spinach

P: 50g C: 30g F: 5g | ~365 cal

4 MEAL 4 — PRE-WORKOUT

6oz ground turkey (93% lean), 1/3 cup sweet potato, greens

P: 35g C: 22g F: 8g | ~300 cal

5 MEAL 5 — POST-WORKOUT

1.5 scoops whey protein, 8oz Oikos Triple Zero yogurt

P: 42g C: 25g F: 3g | ~295 cal

6 MEAL 6 — DINNER

6oz tilapia or lean ground beef (93%), vegetables, hot sauce

P: 32g C: 10g F: 10g | ~260 cal

RESEARCH: METABOLIC ADAPTATION SCIENCE

Metabolic adaptation (adaptive thermogenesis) is a disproportionate reduction in resting energy expenditure beyond what tissue loss alone explains. Jorgensen et al. (2018, Br J Nutr) showed RMR suppression begins at $\geq 5\%$ body weight loss. Phased caloric reduction counters this by preventing the body from fully calibrating to a single deficit level. Fleming et al. (2020, J Funct Morphol Kinesiol) found 2-day carbohydrate refeeds preserved FFM (-0.4 kg vs -1.3 kg) and RMR better than continuous restriction. Incorporating bi-weekly refeeds during Phase 2 is evidence-supported for both metabolic and psychological benefit.

PHASE 3 — FINISHING STRIKE

WEEK 13 | 1,800 KCAL/DAY

PROTEIN 200g (44%)

CARBS 140g (31%)

FAT 50g (25%)

FINAL WEEK RULES

- Zero deviation from plan — every gram counted, every meal timed
- Water intake: 1.5 gallons/day to support final flush
- No alcohol, no cheat meals, no processed food exceptions
- Training volume maintained but intensity monitored — listen to your body
- Sleep 8+ hours minimum — recovery is non-negotiable at this deficit
- This is 7 days. Mental warfare. You finish what you started.

FINAL PUSH MEAL TEMPLATE

~1,790 cal actual

1 MEAL 1 — BREAKFAST

1.5 cups egg whites, 1/4 cup oatmeal (dry)

P: 30g C: 20g F: 3g | ~230 cal

2 MEAL 2 — MID-MORNING

1 scoop whey protein, 1 rice cake

P: 27g C: 14g F: 2g | ~180 cal

3 MEAL 3 — LUNCH

8oz chicken breast, 1/2 cup rice, spinach/broccoli

P: 50g C: 25g F: 5g | ~345 cal

4 MEAL 4 — PRE-WORKOUT

5oz ground turkey (93% lean), 1/4 cup sweet potato, greens

P: 30g C: 16g F: 7g | ~250 cal

5 MEAL 5 — POST-WORKOUT

1.5 scoops whey protein, 6oz Oikos Triple Zero

P: 38g C: 18g F: 2g | ~240 cal

6 MEAL 6 — DINNER

6oz tilapia, large salad with lemon, vegetables

P: 30g C: 8g F: 8g | ~225 cal

RESEARCH: THE 1-WEEK FINISHING STRIKE — BACKED BY SCIENCE

This 1-week aggressive push is scientifically justified — but ONLY for 1 week. At ~33% of TDEE deficit, it remains above the 40% threshold where Carbone & Pasiakos (2019, Nutrients) showed amino acid diversion accelerates markedly. Janssen et al. (2023, Curr Opin Clin Nutr Metab Care) confirmed that short aggressive deficits carry less metabolic damage than extended ones. Aragon et al. (2021, BMC Sports) recommend carbohydrate loading in the final 36-48 hours before a photo shoot to maximize glycogen fullness. This is a finishing strike, not a lifestyle.

WEEKLY SHOPPING LIST

PHASE 1 BASE — ADJUST QUANTITIES PER PHASE

PROTEIN SOURCES

Chicken Breast	5 lbs
Ground Turkey 93%	3 lbs
Flank Steak	1.5 lbs
Tilapia Fillets	2 lbs
Lean Ground Beef 93%	2 lbs
Eggs (whole + whites)	2 doz + 2 cartons
Whey Protein Powder	5 lb tub
Oikos Triple Zero	7 cups
Pure Protein Bars	1 box (12ct)
Beef Jerky	2 bags
MegaFit Meals	3-5 prepped

CARBS & SNACKS

Oatmeal (old fashioned)	1 canister
White Rice (dry)	3 lbs
Bananas	7 (1/day)
Apples (Granny Smith)	7
Pretzels (low-fat)	1 bag
Sweet Potatoes	4 medium
Rice Cakes	1 bag

ESSENTIALS & FLAVOR

Water	1 gal/day min
Mrs. Dash Seasonings	2-3 varieties
Hot Sauce (Frank's)	1 bottle
Cooking Spray (PAM)	1 can
Broccoli (fresh/frozen)	3-4 bags
Spinach (fresh/frozen)	2-3 bags
Lemons / Limes	4-5
Garlic (fresh/powder)	1 each
Low-sodium Soy Sauce	1 bottle
Apple Cider Vinegar	1 bottle
Mustard (yellow/Dijon)	1 bottle

PHASE ADJUSTMENTS

- PHASE 2 (Wk 9-12): Reduce rice by 1/3, cut 1 banana/day, reduce oatmeal portion. Drop almonds/nut butter.
- PHASE 3 (Wk 13): Minimize rice further, no pretzels, no Pure Protein bars. Whole foods only. Eliminate cooking oils.
- ALL PHASES: Maintain full protein purchases. Never cut protein sources to save macros — cut carbs and fats first.

WEEKLY BUDGET TIP: Buy chicken and turkey in bulk. Freeze in pre-portioned bags. Rice and oats in bulk saves 30-40%.

MEAL PREP STRATEGY

COOK ONCE. EAT CLEAN ALL WEEK.

SUNDAY PREP DAY PROTOCOL

PROTEINS (Batch Cook)

- Grill or bake 5 lbs chicken breast — season with Mrs. Dash, garlic, lemon
- Brown 3 lbs ground turkey with onion and garlic — portion into 6oz containers
- Bake tilapia fillets — store individually wrapped
- Hard-boil 1 dozen eggs — keep shelled in fridge for quick access
- Separate egg whites into daily portions (1.5 cup measures)

CARBS (Batch Cook)

- Cook 3 cups dry rice in rice cooker — portion into 1/2 and 2/3 cup servings
- Bake 4 sweet potatoes — cool, peel, portion
- Pre-measure oatmeal into daily bags (1/2 cup or 1/3 cup by phase)

VEGETABLES (Wash & Portion)

- Wash and chop broccoli into daily portions
- Wash spinach, store in paper towel-lined containers
- Slice lemons/limes for the week

CONTAINERS & TRACKING

- Use labeled containers: Meal 1 through Meal 6 for each day
- Pre-log all meals in MyFitnessPal or similar tracker
- Prep 3-4 days at a time for maximum freshness (repeat Wednesday)
- Keep emergency protein bars and jerky in your bag at all times

PRO TIPS

- Invest in a food scale. Eyeballing portions kills cuts. Weigh everything raw.
- Season aggressively — the diet is strict, but flavors don't have to be bland. Zero-calorie sauces are your weapon.
- Cook proteins to 165°F internal. Overcooked chicken = compliance killer. Use a thermometer.
- Frozen vegetables are nutritionally equivalent to fresh and last all week. No excuses.
- If you fail to prep, you will fail the cut. This is the unsexy work that builds the physique.

MEGAFIT MEALS

YOUR GO-TO BACKUP WHEN YOU CAN'T COOK

WHY MEGAFIT MEALS

Life happens. Travel, overtime, emergencies — sometimes you can't meal prep. The difference between a successful cut and a failed one is what you do when your plan breaks. MegaFit Meals is a USDA-approved fitness meal delivery service built by ex-bodybuilders. Every meal is cooked fresh with grass-fed proteins, no preservatives, no GMOs, and flash-frozen for delivery. This is not an excuse to skip prep — this is your safety net when you can't.

RESEARCH: WHY MEAL DELIVERY WORKS FOR CUTTING

Whitham et al. (2013, J Hum Nutr Diet) found meal provision produced 6.6% weight loss vs. 4.3% for self-directed dieting in an RCT — with only 7% dropout vs. 41% dropout in the self-directed arm. Piernas et al. (2019, Obes Rev) meta-analysis showed meal provision programs produce -6.13 kg greater weight loss vs. comparator diets. Ducrot et al. (2017, Int J Behav Nutr Phys Act; n=40,554) found meal planning independently associated with 17% lower obesity odds. Decision fatigue around food is the #1 reason cutting diets fail (Voorhees et al., 2023) — MegaFit eliminates this entirely.

RECOMMENDED MEGAFIT MEALS FOR CUTTING

MEAL	CAL	PROTEIN	CARBS	FAT	BEST FOR
The Go-To Meal	400	41g	47g	4g	Lunch / Dinner
Salmon & Rice	520	36g	62g	18g	Post-Workout
Carne Asada Bowl	530	43g	51g	10g	Lunch (Phase 1)
Sweet & Spicy Chicken	430	40g	56g	4.5g	Pre-Workout
Hibachi Steak Bowl	510	40g	64g	10g	Lunch (Phase 1)

HOW TO USE MEGAFIT IN YOUR CUT

- Order 3-5 meals per week as your emergency backup — store frozen. Microwave in 3-4 minutes.
- PHASE 1: Any meal fits. The Go-To Meal (400 cal, 41g P) and Sweet & Spicy Chicken (430 cal, 40g P) are ideal.
- PHASE 2: Stick to lower-calorie options. The Go-To Meal and Sweet & Spicy Chicken stay within macro targets.
- PHASE 3: Use only The Go-To Meal (400 cal) — lowest calorie option with highest protein-to-calorie ratio.
- Post-Workout: Salmon & Rice (520 cal, 36g P, 62g C) delivers the glycogen replenishment your muscles need.
- NEVER use MegaFit as your primary food source. This is a backup for 2-3 meals per week maximum.
- Always check the label — custom meals can be adjusted for exact macro targets on the MegaFit website.

KEY STAT: Meal provision reduces diet dropout by 5.4x (7% vs 41% attrition). Whitham et al., 2013.

SUPPLEMENT STACK

RESEARCH-BACKED TIMING PROTOCOL — PAGE 1 OF 3

MORNING FASTED

NAC (N-Acetyl Cysteine)

Dose: 600mg (dose 1 of 2)

Timing: Fasted, upon waking with water

WHY: Glutathione precursor and liver/kidney shield. Split dosing sustains plasma levels (t1/2 ~6.25 hrs). Pedre et al. (2021, Pharmacol Ther) reviewed NAC's established role as a GSH precursor and antioxidant, with emerging evidence for H2S-mediated protective signaling. Fasted absorption avoids amino acid competition.

RESEARCH: NAC reduces oxidative stress markers by 30-50% in exercise studies. Synergistic with TUDCA for dual-pathway organ protection (oxidative stress + ER stress). Synergistic with Omega-3 for NF-kB inhibition.

Water + Electrolytes

Dose: 16-20oz water + sodium/potassium

Timing: Immediately upon waking

WHY: Overnight dehydration impairs morning cortisol clearance and cognitive function. Electrolytes support training performance during caloric deficit. Target ~3.7L (1 gallon) total daily intake per ACSM guidelines (Sawka et al., 2007).

RESEARCH: Even 2% dehydration reduces strength output by 10-20%. Sodium supports plasma volume during glycogen-depleted training states. Potassium prevents cramping during high-rep sets.

WITH BREAKFAST (Fat-Containing Meal)

Animal Pak (Multivitamin Complex)

Dose: 1 pack

Timing: With breakfast

WHY: Full-spectrum micronutrient insurance during caloric restriction. Contains zinc (testosterone cofactor at ~15mg), B-complex (B1, B2, B6, B12 for energy metabolism), iron (oxygen transport), and antioxidants (C, E). Deficiency risk increases as food variety narrows during a cut.

RESEARCH: Zinc deficiency alone can reduce testosterone by up to 50% (Prasad et al., 1996). B6 supports protein metabolism. Iron needs increase with intense training. The pack format ensures co-factor synergy.

TUDCA (Tauroursodeoxycholic Acid)

Dose: 500mg

Timing: With breakfast (requires food)

WHY: Chemical chaperone reducing ER stress via GRP78 suppression. Song et al. (2023, Clin Sci) confirmed dual FXR/Nrf2 activation for hepatoprotection. Bile acid analog — must be taken with food. Synergistic with NAC.

RESEARCH: TUDCA improves bile flow and reduces hepatocyte apoptosis. Critical stack component when using Fadogia Agrestis or any supplement with hepatotoxic potential. Reduces ER stress markers within 2-4 weeks of use.

Astragalus Root

Dose: 550mg

Timing: With breakfast

WHY: Astragaloside IV activates PI3K/AKT/mTOR and AMPK pathways. Yeh et al. (2022, Nutrients) confirmed PI3K-mediated Akt/mTOR signaling phosphorylation enhancing myotube hypertrophy. Anti-inflammatory via NF-kB/NLRP3 inhibition, with immunomodulatory and telomerase-activating properties.

RESEARCH: Astragalus has over 2,000 years of use in traditional Chinese medicine. Modern research confirms telomerase activation via TA-65 compound, which may slow cellular aging. The NF-kB and NLRP3 inhibition provides anti-inflammatory protection that complements NAC's oxidative stress pathway. Synergistic with the full organ-protection trio (NAC + TUDCA + Astragalus).

SUPPLEMENT STACK (CONTINUED)

RESEARCH-BACKED TIMING PROTOCOL — PAGE 2 OF 3

WITH BREAKFAST (Continued)

Turkesterone + Tongkat Ali

Dose: 800mg Turk + 400mg TA + 10mg BioPerine

Timing: With breakfast

WHY: Turkesterone acts via ERbeta/mTOR, not androgen receptor. Note: Isenmann et al. (2019, Arch Toxicol) studied ecdysterone specifically — turkesterone effects are extrapolated from structural similarity. Tongkat Ali reduces SHBG and stimulates Leydig cells (Chinnappan et al., 2021, Food Nutr Res; 105-subject RCT). BioPerine inhibits P-gp and CYP enzymes, enhancing absorption of both compounds.

RESEARCH: Tongkat Ali's 105-subject RCT showed statistically significant free testosterone elevation by Week 4-6. BioPerine increases curcumin absorption by 2000% and CoQ10 absorption by 30% via CYP3A4/P-gp inhibition.

Fadogia Agrestis [CYCLE: 5 on / 2 off]

Dose: 600mg (10:1 extract)

Timing: With breakfast (cycling required)

WHY: Saponins stimulate LH-mediated testosterone synthesis (Yakubu et al., 2005, Asian J Androl). NO human RCT data exists — all evidence is from rat models. Nephrotoxicity documented at higher doses (Yakubu et al., 2009). NAC + TUDCA pairing is CRITICAL to counteract membrane stress. Monitor kidney/liver panels quarterly.

UC-II Collagen + Aquamin

Dose: 40mg UC-II + 250mg Aquamin

Timing: With breakfast (light protein preferred)

WHY: UC-II works via oral tolerance induction — Tregs suppress joint inflammation. Schon et al. (2022, JICM) confirmed 40mg/day improved knee ROM in 96-subject RCT. Aquamin provides 70+ trace minerals with independent anti-inflammatory effects (NF-kB/PGE2 inhibition). Must remain undenatured.

RESEARCH: Aquamin's multi-mineral complex includes calcium, magnesium, and 72 trace minerals from marine algae. UC-II must not be heated or mixed with acidic beverages — the 3D collagen structure must remain intact for the oral tolerance immune mechanism to function.

CoQ10 (Ubiquinol)

Dose: 200mg

Timing: With breakfast (fat required for absorption)

WHY: Electron transport chain carrier essential for ATP synthesis. Absorption is enhanced by dietary fat (StatPearls, 'Coenzyme Q10,' NBK531491: absorption improves in the presence of fatty meals). Synergistic with NAC — NAC reduces ROS disruption of the ETC while CoQ10 serves as the direct electron carrier.

RESEARCH: Ubiquinol form is 3-6x more bioavailable than ubiquinone. Supports cellular energy production in every mitochondria-dense tissue — heart, liver, skeletal muscle. Critical during caloric restriction when energy substrates are limited.

Vitamin D3 + K2 (MK-7)

Dose: 5,000 IU D3 + 100mcg K2

Timing: With breakfast (fat-soluble — needs 15-20g dietary fat)

WHY: Raimundo et al. (2011, Int J Endocrinol) showed 60% better D3 absorption with a high-fat meal. K2 directs calcium to bones via osteocalcin carboxylation. Both support testosterone production. D3 + K2 + Aquamin forms a complete bone mineral triad.

RESEARCH: Over 40% of US adults are D3-deficient. Deficiency correlates with lower testosterone, impaired recovery, and increased injury risk. K2 as MK-7 has a 72-hour half-life vs. MK-4's 6-hour half-life, requiring only one daily dose. Without K2, supplemental D3 may direct calcium to arterial walls instead of bone matrix.

SUPPLEMENT STACK (CONTINUED)

RESEARCH-BACKED TIMING PROTOCOL — PAGE 3 OF 3

WITH BREAKFAST (Continued)

Omega-3 Fish Oil (EPA + DHA)

Dose: 2-3g combined EPA/DHA (dose 1 of 2)

Timing: With breakfast

WHY: Anti-inflammatory via prostaglandin/leukotriene modulation. Omega-3 absorption, especially ethyl ester form, is significantly enhanced with dietary fat (Lawson & Hughes, 1988, Biochem Biophys Res Commun). Triglyceride form (TG) absorbs better than ethyl ester (EE). Synergistic with NAC for NF-kB inhibition.

RESEARCH: EPA reduces inflammation; DHA supports neural function and mood stability during caloric restriction. Combined daily EPA+DHA of 4-6g provides maximal anti-inflammatory benefit. Look for IFOS 5-star certified brands.

MORNING + PRE-WORKOUT

iSatori Bio-Gro (Bovine Colostrum Peptides)

Dose: 1.5-3g (1-2 scoops) mixed into shake or water

Timing: 1 scoop with morning shake + 1 scoop pre-workout (15-30 min before training)

WHY: Bio-Pro Bio-Active Peptides from micro-concentrated bovine colostrum. Contains proline-rich peptides (PRPs), immunoglobulins (IgG, IgA, IgM), lactoferrin, growth factors (IGF-1, TGF-beta, FGF). Davison (2021, Nutrients) reviewed 25+ years of BC research: lean mass gains of +1.49kg at 20g/day (Antonio et al.), sprint performance +2.3% vs. whey. Low-dose trials (3.2g/day; Mikellidi et al., 2017, Eur J Nutr) showed recovery and anti-inflammatory benefits comparable to Bio-Gro's 1.5-3g range. Reduces exercise-induced gut permeability by up to 70% (Griffiths et al., 2022, PLOS ONE). Safe with no adverse events in meta-analysis of 10 RCTs (Durkalec-Michalski et al., 2020).

RESEARCH: FOR MEN 18+ WITH 2+ YEARS TRAINING ONLY. BC does NOT raise serum IGF-1 (Jones et al., 2019, Eur J Nutr) — growth factors act locally on GI epithelium and muscle tissue, not systemically. No WADA-relevant concerns. Immunoglobulins support gut integrity during caloric restriction when GI stress from hard training is amplified. Mix unflavored powder into any protein shake. MCT oil base enhances absorption.

PRE / INTRA WORKOUT

Optimum Nutrition Amino Energy

Dose: 2 scoops (10g amino blend + 200mg caffeine + 100mg green tea extract)

Timing: 15-30 min pre-workout or sip intra-workout

WHY: Full-spectrum amino blend: micronized taurine, L-glutamine, L-arginine, L-leucine, beta-alanine, L-citrulline, L-valine, L-tyrosine, L-histidine, L-lysine HCl, L-phenylalanine, L-threonine, L-methionine. Leucine triggers mTOR/MPS signaling (Churchward-Venne et al., 2012, Nutr Metab). Leucine-enriched EAAs during exercise boost post-exercise MPS by 33% (Pasiakos et al., 2011, Am J Clin Nutr). Beta-alanine buffers muscle pH for extended high-intensity sets (ISSN Position Stand, Trexler et al., 2015; effect size 0.39 in trained males, Georgiou et al., 2024). Citrulline enhances NO production for vasodilation and blood flow (Kim et al., 2023, Nutrients). Taurine extends time to exhaustion by 10% and reduces lactate by 16.5% (Page et al., 2019, Eur J Sport Sci).

RESEARCH: CAFFEINE + GREEN TEA SYNERGY: 100mg caffeine + 50mg green tea extract per scoop. Dulloo et al. (1999, Am J Clin Nutr) landmark RCT showed caffeine + catechins increase fat oxidation 28-41% via COMT inhibition. Roberts et al. (2015, JISSN) confirmed 24.9% fat oxidation increase and 1.63% body fat reduction. ISSN Position Stand (Grgic et al., 2021): caffeine improves strength, power, and endurance. Combined taurine + BCAA reduces DOMS (Ra et al., 2013, JISSN). Particularly important during Phases 2-3 when glycogen is depleted and training intensity must be maintained on reduced calories.

SUPPLEMENT STACK (EVENING)

DINNER & BEFORE BED PROTOCOL

WITH DINNER

NAC 600mg (2nd dose) + Omega-3 Fish Oil (2nd dose)

Split dosing sustains 24-hour antioxidant coverage and anti-inflammatory EPA/DHA levels.

RESEARCH: NAC half-life is ~6.25 hours. Morning + dinner dosing maintains consistent glutathione precursor availability.

BEFORE BED

Magnesium Glycinate

Dose: 400mg (elemental ~56-80mg Mg)

Timing: 30-60 minutes before bed

WHY: Modulates GABA receptors and NMDA for sleep quality. The glycine chelate provides independent sleep-promoting effects.

Gratwicke et al. (2021, *Nutrients*) reviewed nutritional interventions for athlete sleep, identifying magnesium as evidence-supported for improvement. Bypasses laxative effect of citrate/oxide forms.

RESEARCH: Glycinate form has the highest bioavailability of all magnesium forms. Glycine itself acts as an inhibitory neurotransmitter, providing a dual sleep benefit. Magnesium also supports 300+ enzymatic reactions including protein synthesis and muscle contraction. Deficit is common in athletes due to sweat losses.

ISATORI BIO-GRO — SCIENCE DEEP DIVE

WHAT'S IN IT

- Proline-Rich Peptides (PRPs): Modulate immune cell activity and cytokine production. Support tissue repair signaling.
- Immunoglobulins (IgG, IgA, IgM): Stabilize gut epithelium, prevent pathogen adhesion, reduce exercise-induced gut permeability. IgG concentration in BC: 20-150 g/L (Bicen et al., 2024, *Front Immunol*).
- Lactoferrin: Broad antimicrobial, antiviral, and anti-inflammatory properties. Supports iron metabolism without GI side effects. Concentration in BC: ~1.5 g/L (Riaz et al., 2026, *J Anim Feed Sci*).
- Growth Factors (IGF-1, TGF-beta, FGF): Act locally on GI epithelium and muscle tissue. Do NOT raise systemic IGF-1 levels (Jones et al., 2019). Support tissue repair and anabolic signaling via PI3K/Akt/mTOR (Yoshida & Delafontaine, 2020, *Cells*).
- MCT Oil Base: Medium-chain triglycerides enhance absorption of fat-soluble bioactives. Rapidly oxidized for energy, not stored as fat.

WHO SHOULD USE BIO-GRO

- Men 18+ with at least 2 years of consistent resistance training experience.
- Athletes in caloric deficit who need recovery support and gut health protection.
- Individuals training 4+ days/week who experience slow recovery or frequent illness during cuts.

DOSING PROTOCOL

- Standard: 1 scoop (1.5g) mixed into morning protein shake. Unflavored — no taste impact.
- Enhanced: 2 scoops (3g) on training days — 1 scoop AM, 1 scoop pre-workout (15-30 min before training).
- Low-dose BC trials (3.2g/day) showed significant anti-inflammatory and recovery benefits in trained athletes (Mikellidi et al., 2017; Skarpanska-Stejnborn et al., 2023).

SAFETY NOTE: Contains milk (bovine colostrum). Not for individuals with dairy allergy.

Safe in meta-analysis of 10 RCTs with no adverse events at any dose (Durkalec-Michalski et al., 2020, *Nutrients*).

SUPPLEMENT SCIENCE APPENDIX

INTERACTIONS, CYCLING, ABSORPTION & WARNINGS

STACK SYNERGY MAP

- NAC + TUDCA + Astragalus: Triple-layer organ protection — oxidative stress (NAC), ER stress (TUDCA), immune/inflammatory (Astragalus). Non-overlapping, complementary mechanisms.
- NAC + CoQ10: Mitochondrial synergy — NAC reduces ROS disruption of electron transport; CoQ10 is the direct electron carrier.
- Tongkat Ali + Fadogia: Complementary testosterone pathways — Fadogia via LH elevation, Tongkat Ali via SHBG reduction + Leydig cell support.
- D3 + K2 + Aquamin: Complete bone mineral triad — D3 enhances calcium absorption, K2 directs it to bones, Aquamin provides the multi-mineral matrix.
- BioPerine (from Tongkat): Whole-stack enhancer — inhibits P-gp and CYP enzymes, boosting absorption of Astragalus, Turkesterone, CoQ10, D3, and K2.
- Omega-3 + NAC: Dual anti-inflammatory — NAC via NF-kB inhibition, Omega-3 via prostaglandin/leukotriene modulation. Additive effect.
- Bio-Gro + Amino Energy (Pre-Workout Stack): Bio-Gro primes gut integrity and delivers growth factors; Amino Energy provides MPS substrates, caffeine, and performance enhancement. Take both 15-30 min before training.
- Amino Energy Caffeine + Green Tea: Synergistic thermogenesis — caffeine elevates catecholamines, catechins inhibit COMT to extend their fat-burning effect (Dulloo et al., 1999).

INTERACTION WARNINGS & ABSORPTION COMPETITION

- Calcium vs. Magnesium: Both Animal Pak and Aquamin provide calcium. Keep breakfast calcium below 500mg to avoid blocking Mg absorption. Take Mg at bedtime to separate timing.
- Iron vs. Calcium: Animal Pak contains iron. Calcium competes for absorption. If iron status is a concern, take Animal Pak separately from Aquamin.
- NAC vs. Dietary Amino Acids: Fasted NAC (morning dose) avoids cysteine competition from protein meals. Dinner NAC with food is fine — pharmacodynamic effect is less PK-dependent.
- Fat-Soluble Vitamin Cluster: D3, K2, CoQ10, and Omega-3 ALL require dietary fat for absorption. Taking them together at breakfast with 15-20g fat maximizes a single absorption window.
- NAC + Nitroglycerin: Do NOT combine. NAC potentiates hypotensive effects. Not relevant to this stack but critical if cardiovascular medications are ever prescribed.
- Bio-Gro + Dairy Allergy: Bio-Gro contains bovine colostrum (milk). Contraindicated for individuals with confirmed dairy/milk protein allergy.

CYCLING PROTOCOLS WITH EVIDENCE

- Fadogia Agrestis: 5 days on / 2 days off (minimum). Add a full 2-week break every 6-8 weeks. Yakubu et al. (2009) documented subclinical membrane disruption reversible with 10-day washout. Nephrotoxicity risk is cumulative — DO NOT skip cycles.
- Turkesterone: No established cycling protocol needed — no evidence of receptor downregulation or hormonal axis suppression. Continuous use is acceptable.
- Tongkat Ali: Can be used continuously. Clinical RCTs (Chinnappan et al., 2021) used continuous daily dosing for 12+ weeks with sustained benefit.
- NAC: Continuous use supported. No cycling required. Long-term safety data up to 1,200mg/day is robust.
- Bio-Gro (Bovine Colostrum): No cycling required. Meta-analysis of 10 RCTs supports continuous daily use with no adverse events (Durkalec-Michalski et al., 2020). Long-term studies up to 24 weeks show sustained benefit.
- Amino Energy: No cycling required for amino acids. Caffeine tolerance may develop — consider 1-week caffeine wash every 8-12 weeks to restore sensitivity.

SUPPLEMENT QUALITY STANDARDS

- Third-Party Testing: Prioritize NSF Certified for Sport, Informed Sport, or USP-verified supplements. These certifications ensure absence of banned substances and label accuracy.
- Omega-3: Look for IFOS 5-star certification. Verify triglyceride (TG) form over ethyl ester (EE) on the label. Molecular distillation removes heavy metals and PCBs.
- CoQ10: Ubiquinol (reduced form) is 3-6x more bioavailable than ubiquinone (oxidized form). Kaneka QH is the gold-standard branded ingredient.
- Colostrum (Bio-Gro): Ensure first-milking sourced (highest IgG concentration). Look for low-heat processing to preserve immunoglobulin structure and bioactivity.
- Tongkat Ali: Standardized root extract (200:1 or higher). Verify eurycomanone content on COA. Avoid proprietary blends that hide actual dosing.
- Storage: Keep all probiotics, colostrum, and omega-3 products refrigerated after opening. Heat and light degrade bioactive compounds rapidly.

ABSORPTION & TIMING GUIDE

MAXIMIZE EVERY SUPPLEMENT DOLLAR

KEY ABSORPTION TIPS

- Fat-soluble vitamins (D3, K2, CoQ10): Require 15-20g dietary fat. Raimundo et al. (2011) showed 60% higher D3 absorption with fat-containing meals.
- NAC timing: Fasted morning for maximum absorption; with-food dinner for renal/hepatic coverage. Split dosing sustains plasma levels (t1/2 ~6.25 hrs).
- TUDCA: MUST be taken with food — bile acid metabolism is food-dependent. Fasted administration reduces efficacy and causes GI discomfort.
- UC-II Collagen: Must remain undenatured. Do not heat, blend with acidic drinks, or store improperly. The oral tolerance mechanism requires intact 3D collagen structure.
- Omega-3: Triglyceride form (TG) absorbs better than ethyl ester (EE). Always take with a fat-containing meal. Never take fasted.
- Bio-Gro: Mix into a protein shake (not hot liquid). The immunoglobulins and growth factors require intact protein structure for bioactivity. MCT oil base aids absorption. Take AM scoop in breakfast shake, pre-workout scoop in water or shake 15-30 min before training.
- Amino Energy: Consume 15-30 minutes before training alongside Bio-Gro scoop 2 for peak plasma amino acid levels and growth factor delivery during exercise.
- BioPerine (in Tongkat Ali): Enhances absorption of CoQ10 by 30%, curcumin by 2000%, and fat-soluble vitamins. Take at the same time as your fat-soluble cluster.

DAILY SUPPLEMENT TIMING SUMMARY

TIME	SUPPLEMENTS	NOTES
Upon Waking	NAC 600mg, Water + Electrolytes	Fasted, before food
Breakfast	Animal Pak, TUDCA 500mg, Astragalus, Turk+TA, Fadogia, UC-II+Aquamin, CoQ10, D3+K2, Omega-3, Bio-Gro	With 15-20g dietary fat All fat-soluble supps here Bio-Gro scoop 1 in shake
Pre-Workout	Bio-Gro (scoop 2) + Amino Energy	15-30 min before training
Dinner	NAC 600mg, Omega-3 (2nd dose)	With food
Before Bed	Magnesium Glycinate 400mg	30-60 min before sleep

"The supplement stack is the force multiplier. The diet is the foundation. Never invert this."

MONTHLY SUPPLEMENT COST ESTIMATE

SUPPLEMENT	30-DAY SUPPLY	DAILY COST
Animal Pak	\$28-35	~\$1.00
NAC 600mg (x2/day)	\$12-18	~\$0.50
TUDCA 500mg	\$25-35	~\$1.00
Astragalus 550mg	\$10-15	~\$0.40
Turkesterone 800mg	\$35-50	~\$1.50
Tongkat Ali 400mg + BioPerine	\$20-30	~\$0.85
Fadogia Agrestis 600mg	\$18-25	~\$0.70
UC-II 40mg + Aquamin	\$20-28	~\$0.80
CoQ10 (Ubiquinol) 200mg	\$25-40	~\$1.10
D3 5000IU + K2 100mcg	\$12-18	~\$0.50
Omega-3 (IFOS rated)	\$20-30	~\$0.85
Magnesium Glycinate 400mg	\$12-18	~\$0.50
Bio-Gro (1-2 scoops/day)	\$30-40	~\$1.10
Amino Energy (2 scoops/day)	\$22-30	~\$0.90
TOTAL ESTIMATED	\$290-410/mo	~\$11.70/day

Prices reflect average US retail as of 2025. Buy in bulk or subscribe-and-save to reduce cost by 15-25%.

TIMELINE TO RESULTS

13-WEEK MILESTONES — EVIDENCE-BASED PROJECTIONS

Starting point: ~200 lbs, ~15-18% body fat | Target: 10-12% body fat | Expected loss: 13-18 lbs (85-90% fat mass)

WEEKS 1-4 ESTABLISHING THE DEFICIT

- Scale drops 2-4 lbs rapidly in Week 1 — primarily glycogen depletion and water, not fat
- True fat loss accumulates from Week 2 onward at 0.5-1 lb/week
- Waist measurement decreases 0.5-1 inch by Week 4
- Training performance may fluctuate slightly as body adjusts to new energy intake
- Supplement stack begins organ-protective effects (NAC: GSH elevation in 24-72 hrs; TUDCA: ER stress reduction by Week 2-4)
- Bio-Gro gut-protective effects establish within first 2 weeks — reduced exercise-induced permeability

WEEKS 5-8 VISIBLE CHANGES BEGIN

- Consistent fat loss of 0.5-1 lb/week becomes visually apparent
- Vascularity in forearms and upper arms begins to appear or improve
- Muscle definition in shoulders, chest separation, and upper abs visible at 12-14% BF
- Tongkat Ali testosterone effects measurable by Week 4-6 (Chinnappan et al., 2021)
- UC-II joint benefits begin at Week 3-4 and continue improving
- Amino Energy caffeine + green tea synergy maximizing fat oxidation during training sessions
- Total weight lost by Week 8: approximately 6-10 lbs of predominantly fat mass

WEEKS 9-12 SIGNIFICANT RECOMPOSITION

- Approaching 10-12% body fat — visible abs (top 4) and oblique striations emerge
- Phase 2 caloric reduction (2,000 cal) may cause slight performance decrements — normal
- Metabolic adaptation now measurable — implement 2-day carb refeeds every 10-14 days
- Hormonal markers (testosterone, T3) may begin suppressing (Chappell et al., 2021)
- Supplement stack at full protective capacity: NAC/TUDCA shielding, CoQ10 energy, Mg sleep quality

WEEK 13 THE FINISHING STRIKE

- Final aesthetic refinement at 1,800 kcal — revealing existing work, not building new tissue
- Subcutaneous water redistributes, veins become more prominent
- Consider carbohydrate loading in final 36-48 hours for peak glycogen fullness (Aragon et al., 2021)
- Total 13-week result: 13-18 lbs lost, final body fat 8-12%, lean mass preserved within 1-3 lbs

POST-CUT: REVERSE DIETING PROTOCOL

- Increase calories by 75-150 kcal per week from end-of-cut intake (1,800) back toward TDEE (~2,700-2,800)
- Prioritize carbohydrate increases first — restores leptin, thyroid (T3), and training performance
- Maintain high protein throughout (200g+) to protect lean mass during metabolic recovery
- Continue full supplement stack during reverse — NAC/TUDCA organ protection remains critical during metabolic transition
- Expect 4-8 weeks to fully restore resting energy expenditure (Chica-Latorre et al., 2022, JISSN). Note: reverse dieting remains a theoretical concept with limited controlled evidence.
- Fat overshoot phenomenon is real — structured reverse dieting aims to prevent rebound (Melby et al., 2017)
- Monitor bodyweight weekly and waist measurements biweekly — accept 2-4 lbs of water/glycogen regain as normal

"The body achieves what the mind believes."

Trust the process. Execute the plan. Results are inevitable.

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PEPTIDE PROTOCOL

Dosing // Timing // Injection Technique // Bloodwork // Dosing Calculator



PEPTIDE PROTOCOL

RESEARCH & EDUCATION GUIDE

Dosing // Timing // Benefits // Injection Protocol

Built on Real-World Application & Bloodwork Oversight

Written by Someone Running the Stack.

PREREQUISITE: 2+ YEARS CONSISTENT TRAINING

This guide assumes a foundation of discipline, nutrition, and consistent resistance training.
If you haven't earned the base, these tools are not for you yet.

[FOR RESEARCH & EDUCATIONAL PURPOSES ONLY]

THIS IS NOT HEALTH ADVICE. THIS IS NOT MEDICAL ADVICE.

All compounds referenced are research compounds. The author is not a medical professional.

You **MUST** consult a licensed physician before beginning any peptide protocol.

All dosing reflects published research protocols and personal application, not clinical prescriptions.

MOST IMPORTANTLY: CONSULT YOUR DOCTOR.

◆ **BACKED BY 56 PEER-REVIEWED REFERENCES** ◆

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FROM THE FOUNDER

"Seventeen years under the bar. Not seventeen months. Seventeen years."

I started training at 14 years old. By 17, I was benching 300 lbs at 175 lbs bodyweight. Not because I had good genetics — because I didn't miss days. I built to a 405 bench press and 225x27 reps at 200 lbs bodyweight, natural, over nearly two decades of relentless consistency.

I didn't touch a peptide until I had exhausted every ounce of natural potential. Every rep, every meal, every hour of sleep was accounted for long before I ever reconstituted a vial.

"Peptides are tools. A tool in the hands of someone who hasn't built the foundation is a liability."

This guide exists because I got tired of watching people skip the work. They want the shortcut before they've earned the right to optimize. Peptides don't replace discipline — they enhance it. If your training isn't dialed, your nutrition isn't locked, and your sleep isn't prioritized, peptides will do nothing for you except drain your wallet.

Everything in this document is based on my own research, my own bloodwork, my own application, and the published science that backs it. I'm not a doctor. I'm not giving you medical advice. I'm telling you what I run, why I run it, and what the research says.

If you're here, I'm assuming you've put in the work. If you haven't — close this document and go train.

— **Leonidas Lifts**

THE FOUNDATION

17+ Years Natural Training | Bench Press: 405 lbs | 225 x 27 Reps at 200 BW
5+ Days/Week Training | 200g+ Protein Daily | Bloodwork Every 90 Days

Built the Foundation First. Then Optimized.

PREREQUISITES & SUPPLIES

■ CONSULT YOUR DOCTOR BEFORE STARTING ANY PEPTIDE PROTOCOL ■

You must consult a licensed physician before beginning any peptide protocol. This document is for research and educational purposes only. It is **NOT** health advice, medical advice, or a prescription. Do not self-administer any compound without direct physician oversight and baseline bloodwork.

DO NOT USE PEPTIDES IF

- ✗ You have NOT trained 5+ days/week consistently for at least 2 years
- ✗ You have NOT completed baseline bloodwork with a licensed physician
- ✗ You are under 21 years of age
- ✗ You have NOT consulted a physician about peptide use
- ✗ You are unwilling to monitor bloodwork on-cycle (mandatory)

RECONSTITUTION STANDARD

Standard: Add **300 units (3mL) of bacteriostatic (BAC) water** to each peptide vial. Draw slowly down the side of the vial — never spray directly onto the lyophilized powder. Swirl gently. Never shake. Refrigerate after reconstitution. Vials are typically stable for 28-30 days refrigerated.

Dosing math: With 300 units BAC water in a standard 5mg vial: 10 units on an insulin syringe = ~167 mcg. 20 units = ~333 mcg. 25 units = ~417 mcg. Adjust based on your vial concentration.

INJECTION SUPPLIES CHECKLIST

- ✓ **Insulin Syringes** — 29g or 31g, 0.5mL or 1mL — for SubQ injection
- ✓ **Bacteriostatic Water** — 30mL vial — for reconstitution (NOT sterile water)
- ✓ **Alcohol Swabs (CRITICAL)** — Use a FRESH swab before EVERY vial draw AND EVERY injection. Never reuse swabs.
- ✓ **Sharps Container** — Safe disposal — never reuse or recap needles
- ✓ **Peptide Storage** — Refrigerate all reconstituted vials at 36-46°F (2-8°C)
- ✓ **Gauze Pads / Cotton Balls** — For gentle pressure post-injection — do not rub the site
- ✓ **Clean Flat Workspace** — Sanitize your workspace surface before every injection session

STERILITY IS NON-NEGOTIABLE

Every injection carries a risk of contamination if sterility protocols are not followed to the letter. Bacterial infection, abscess formation, and systemic sepsis are real consequences of cutting corners. The **alcohol swab protocol** outlined on the next page is your primary defense. Follow it without exception, every single time.

The number one mistake beginners make: **reusing alcohol swabs** or skipping the vial swab when drawing from the same vial twice in one session. Every surface gets a fresh swab. No exceptions. One swab, one surface, one time.

KEY RULE: WHEN IN DOUBT, SWAB AGAIN.

If you cannot remember whether you swabbed the vial stopper — swab it again with a fresh pad.
If you cannot remember whether you swabbed the injection site — swab it again with a fresh pad.

Alcohol swabs cost fractions of a penny. Infections cost thousands of dollars and weeks of recovery.

ALCOHOL SWAB PROTOCOL — MANDATORY FOR EVERY INJECTION

Sterility is **non-negotiable**. Contamination at any point in the injection process can lead to infection, abscess formation, or systemic complications. Alcohol swabs (70% isopropyl alcohol prep pads) are your primary defense against bacterial contamination.

WHEN TO SWAB — EVERY SINGLE TIME:

■ SWAB THE VIAL STOPPER

Before EVERY draw from ANY vial. Even if you just swabbed it 30 seconds ago for a different peptide — swab it again. Use a fresh swab each time. Wipe firmly across the rubber stopper in one direction.

■ SWAB THE INJECTION SITE

Before EVERY injection. Use a NEW, fresh alcohol swab — never reuse the one you used on the vial. Wipe the skin in a circular motion, starting from the center and moving outward in a 2-3 inch radius.

■ LET IT DRY — 30 SECONDS

After swabbing the injection site, allow the alcohol to air dry for a full 30 seconds before inserting the needle. Injecting through wet alcohol causes stinging and can carry surface bacteria into the tissue. Do NOT blow on it or fan it dry.

■ NEVER REUSE A SWAB

Each alcohol swab is single-use. One swab per surface. If you are drawing from two vials (e.g., CJC-1295 and Ipamorelin), that is two separate swabs for the vial stoppers plus one swab for the injection site — minimum three swabs per session.

SUBCUTANEOUS (SUBQ) INJECTION TECHNIQUE — STEP BY STEP

1. Wash hands thoroughly with soap and warm water for at least 20 seconds. Dry with a clean towel.
2. Swab the vial stopper with a **fresh alcohol swab**. Wipe firmly in one direction. Allow to air dry.
3. Draw the prescribed dose from the vial using a new insulin syringe. If drawing from multiple vials, use a **fresh swab on each vial stopper** before each draw.
4. Swab the injection site with a **new, unused alcohol swab**. Circular motion, center outward, 2-3 inch radius.
5. **Wait 30 seconds** for the alcohol to fully air dry before proceeding. Do not blow on it.
6. Pinch a fold of skin at the abdomen (around the navel, 2 inches away) or upper thigh.
7. Insert needle at a 45-90 degree angle into the pinched skin fold.

8. Inject slowly and steadily. Withdraw needle. Apply gentle pressure with a clean gauze pad — do not rub.

9. Dispose of the needle immediately in your sharps container. **Never recap or reuse needles.** One needle, one injection.

10. Rotate injection sites with every injection to prevent lipodystrophy (tissue hardening or fat loss at site).

SWAB COUNT PER SESSION EXAMPLE

AM injection (CJC-1295 + Ipamorelin combined): Swab CJC vial (1) + Swab Ipa vial (2) + Swab injection site (3) = **3 swabs minimum**

Pre-workout injection (BPC-157): Swab BPC vial (1) + Swab injection site (2) = **2 swabs minimum**

PM injection (CJC-1295 + Ipamorelin): Swab CJC vial (1) + Swab Ipa vial (2) + Swab injection site (3) = **3 swabs minimum**

Stock at least 200-300 alcohol swabs per month. Buy in bulk. Never run out. No swabs = no injection.

POST-INJECTION CARE & SITE ROTATION

After each injection, apply gentle pressure with a clean gauze pad for 5-10 seconds. **Do not rub the injection site** — rubbing can cause the peptide to disperse from the subcutaneous depot and may increase bruising.

Site rotation is critical for long-term use. Repeated injections in the same location can cause lipodystrophy — tissue hardening, fat atrophy, or lumps under the skin. Use at least 4-6 different injection sites and rotate systematically:

- ✓ Left abdomen (2 inches from navel)
- ✓ Right abdomen (2 inches from navel)
- ✓ Left upper thigh (anterior)
- ✓ Right upper thigh (anterior)
- ✓ Left love handle area (lateral abdomen)
- ✓ Right love handle area (lateral abdomen)

Keep a simple log (notebook or phone) to track which site you used for each injection. This prevents accidentally using the same site repeatedly. If you notice any redness, swelling, warmth, or pain lasting more than 48 hours at an injection site, **stop injecting at that site and consult your physician immediately.**

CORE PEPTIDE STACK

Growth Hormone Axis + Gene Reprogramming

CJC-1295 (NO DAC) + IPAMORELIN — GH SECRETAGOGUE COMBO

Dose	Frequency	Timing	Fasting
20 units each	2x daily	AM fasted + PM before bed	90 min window (no food)

BENEFITS: Deep sleep enhancement (SWS), increased GH pulsatility, fat oxidation, lean mass preservation, collagen synthesis, improved recovery. IGF-1 elevation of 50-150+ ng/mL above baseline within 4-8 weeks.

THE SCIENCE: CJC-1295 (No DAC) is a modified GHRH analog that activates pituitary GHRHR via cAMP/PKA pathway, priming somatotropes for GH release. Ipamorelin is a selective GHSR-1a agonist that triggers the calcium-mediated GH secretory burst via PLC/IP3 pathway. Combined, they produce synergistic GH pulses exceeding either compound alone — the "extend and trigger" model. Half-lives: CJC No DAC ~30 min; Ipamorelin ~2 hrs (Rasmussen et al., 1999).

KEY STUDIES: Raun et al., Eur J Endocrinol 1998 — Ipamorelin: first selective GH secretagogue; no ACTH/cortisol elevation at 200x effective dose. Rasmussen et al., 1999 — Ipamorelin terminal half-life ~2 hours.

CRITICAL NOTE: The Teichman et al., JCEM 2006 PK data (GH 2-10x for 6+ days; IGF-1 1.5-3x for 9-11 days) describes CJC-1295 **WITH DAC**, not the No-DAC version. The No-DAC version (Mod GRF 1-29) has an approximate half-life of ~30 minutes based on analog pharmacology.

SYNERGY: CJC/Ipa elevates GH/IGF-1 systemically. BPC-157 upregulates GH receptor expression in healing tissues (JAK2/GHR). GHK-Cu drives collagen via TGF-beta independently. TB-500 ILK/Akt pathway is enhanced by IGF-1 co-signaling. All four compounds synergize through distinct, non-redundant mechanisms.

GHK-CU — COPPER PEPTIDE / GENE REPROGRAMMING

Dose	Frequency	Timing	Route
10 units	Daily	Morning	SubQ

BENEFITS: Collagen/elastin synthesis, anti-inflammatory (NFkB suppression), skin/hair regeneration, DNA repair gene activation, antioxidant enzyme upregulation, neuroprotection, scar remodeling.

THE SCIENCE: GHK-Cu is an endogenous tripeptide that modulates expression of ~30% of the analyzed human gene database (~4,000 genes). It activates TGF-beta/SMAD pathway for collagen, suppresses NFkB/p38 MAPK for inflammation, upregulates 41 ubiquitin-proteasome genes for protein quality control, and 47 DNA repair genes. It operates via copper delivery to metalloenzymes (SOD1, lysyl oxidase) and epigenetic transcriptional reprogramming. Plasma half-life: minutes (but transcriptional effects persist).

KEY STUDY: Pickart & Margolina, IJMS 2018 — GHK modulates ~4,000 genes (~30% of analyzed gene database); reverses COPD and cancer metastasis gene signatures via Connectivity Map analysis. Maquart et al., FEBS 1988 — Collagen synthesis stimulation beginning at picomolar

concentrations.

PRACTICAL NOTE: GHK-Cu is commonly applied topically (serums, patches) but SubQ injection delivers systemic gene-modulatory effects that topical cannot replicate. Reconstitute with bacteriostatic water and inject 1-2 mg SubQ daily for cycles of 4-8 weeks. The peptide's copper-binding activity means copper intake should be monitored — excess copper supplementation is not recommended during GHK-Cu cycles.

STACKING CONTEXT: GHK-Cu pairs well with BPC-157 for comprehensive tissue remodeling — BPC-157 drives vascular scaffolding and NO signaling while GHK-Cu handles extracellular matrix (collagen, elastin, decorin) remodeling at the gene expression level. This is the difference between building new blood vessels to an injury site (BPC-157) and rebuilding the structural tissue itself (GHK-Cu).

SOURCING WARNING: GHK-Cu peptide quality varies enormously. The copper ion must be properly chelated to the GHK tripeptide — free copper alone is toxic. Third-party COA (Certificate of Analysis) is mandatory. Verify copper content matches specification (should be ~1.5% by mass). Reputable vendors provide HPLC purity reports and amino acid analysis. If the vial does not have a blue tint after reconstitution, the copper binding may be inadequate.

GH AXIS STACK SUMMARY

CJC-1295/Ipamorelin: 20 units each, 2x daily (AM fasted + PM before bed). Primary GH/IGF-1 elevator.

GHK-Cu: 10 units daily (AM). Gene reprogramming + collagen synthesis + DNA repair.

MK-677: 12.5–25 mg oral at bedtime (if used). Complementary to CJC/Ipa but watch glucose.

These three compounds address the growth hormone axis from different angles: CJC/Ipa drives pulsatile GH release, GHK-Cu handles downstream tissue remodeling at the gene expression level, and MK-677 (optional) provides sustained GH elevation via ghrelin mimicry. Together they create the most comprehensive natural GH optimization available.

"Growth hormone is the body's master repair signal. These compounds don't replace it — they amplify what your training already triggers."

IMPORTANT TIMING NOTE: CJC-1295/Ipamorelin should be injected on an empty stomach — food (especially carbohydrates and fats) blunts GH release via insulin and somatostatin antagonism. The 90-minute fasting window pre- and post-injection is not optional. This is the single most common dosing mistake. If you eat within 90 minutes of injection, you are significantly reducing the GH pulse amplitude and defeating the purpose of the protocol.

CORE PEPTIDE STACK

Tissue Repair + Cellular Migration

BPC-157 — BODY PROTECTION COMPOUND

Dose	Frequency	Timing	Route
25 units	Every other day (EOD)	45 min pre-workout	SubQ (near injury/abdomen)

BENEFITS: Tendon/ligament/muscle repair, gut healing, angiogenesis, GH receptor upregulation, NSAID damage reversal, connective tissue remodeling, anti-inflammatory (COX-NO axis).

THE SCIENCE: BPC-157 activates the Src-Caveolin-1-eNOS pathway (NO generation), FAK-paxillin (fibroblast adhesion), JAK2/GHR (GH receptor upregulation), and a rapid early gene cascade (Akt1, VEGFA, EGFR, MAPK within 2-10 min). It upregulates egr-1/nab2 for a self-limiting safe healing signal. Uniquely stable in gastric juice for 24+ hours. Elimination half-life: 5-15 min in animal models (He et al., 2022); tissue activity persists ~102 hours via active metabolites.

KEY STUDIES: Hsieh et al., Sci Rep 2020 — Src-Cav-1-eNOS pathway; vasorelaxation (16-48%). He et al., Front Pharmacol 2022 — ADME: proposed clinical dose 200 mcg/person/day; t_{1/2} = 15.2 min (rats), 5.27 min (dogs). Vasireddi et al., Sports Health 2025 — Systematic review (36 studies): 7/12 human patients reported >6 months knee pain relief.

SYNERGY: BPC-157 creates the vascular/signaling environment for repair (NO, VEGF, GHR upregulation). TB-500 then drives cellular migration into that environment. Together they address the two rate-limiting steps of healing: signaling scaffold + cell mobilization.

TB-500 (THYMOSIN BETA-4) — CELLULAR MIGRATION

Dose	Frequency	Timing	Route
25 units	EOD	Any time (offset BPC-157 days preferred)	SubQ

BENEFITS: Cell migration/repair, anti-fibrotic (reduces scarring), cardiac progenitor activation, anti-inflammatory (ICAM-1/NFκB suppression), antioxidant (SOD/catalase upregulation), scar remodeling, connective tissue flexibility.

THE SCIENCE: TB-500 binds G-actin monomers (K_d ~0.5 uM), preventing F-actin polymerization and increasing cell migration capacity. Activates ILK/Akt pathway for cell survival. Uniquely re-activates the embryonic epicardial program in adult hearts for cardiac regeneration. The oxidized form (Tb4-SO) inhibits IFN-gamma for inflammation resolution. Plasma half-life data not established in published human PK studies; cytoskeletal and transcriptional effects persist for days.

KEY STUDIES: Peng et al., AJP Heart 2014 — Tb4 prevented cardiac rupture (56% to 23%); improved ejection fraction; reduced ICAM-1 and MMPs. Maar et al., Cells 2021 — ILK/Akt mechanism; epicardial progenitor reactivation. Active Phase 1/2 clinical trial (NCT07487363, TBRIDGE-CV) for stable ASCVD — TB-500 17-23 fragment, started Feb 2026.

RETATRUTIDE

Triple Agonist — GLP-1 / GIP / Glucagon

Mechanism	Frequency	Ramp	Route
GLP-1 + GIP + Glucagon	Once weekly	50 -> 100 units over 6 wks	SubQ

TITRATION RAMP SCHEDULE

Week	Dose	Notes
Week 1-2	50 units (~1.67 mg)	Starting dose — assess GI tolerance
Week 3-4	75 units (~2.5 mg)	Mid-ramp — nausea typically peaks here
Week 5-6+	100 units (~3.33 mg)	Maintenance — hold when GI side effects resolve

THE SCIENCE: Retatrutide simultaneously activates three hormonal pathways. GLP-1: appetite suppression + gastric slowing. GIP: enhanced insulin secretion + reduced GI side effects. Glucagon: increased resting energy expenditure via hepatic fatty acid oxidation + preferential visceral fat reduction. The glucagon arm drives metabolic rate increases absent in semaglutide and tirzepatide. Retatrutide achieved >80% liver fat reduction with 89-93% of participants reaching <5% liver fat.

PHASE 2: Jastreboff et al., NEJM 2023 — 12 mg dose: -17.5% body weight at 24 weeks (-41.2 lbs); -24.2% at 48 weeks (-57.8 lbs). No plateau reached.

PHASE 3 (TRIUMPH-4): 12 mg: -28.7% body weight at 68 weeks (-71.2 lbs); 58.6% achieved ≥25% weight loss. 75.8% reduction in WOMAC knee pain scores. **Citation: Eli Lilly press release, December 2025; NCT05869903; peer-reviewed publication pending.**

ANTI-CANCER DATA: Marathe et al., npj Metabolic Health and Disease, 2025 — 14-fold pancreatic tumor volume reduction (vs. vehicle); 17-fold lung cancer tumor reduction. 3.5x more effective than semaglutide. Anti-tumor immunity durable even after drug withdrawal.

RETATRUTIDE — CONTINUED

COMPARISON: RETATRUTIDE VS SEMAGLUTIDE VS TIRZEPATIDE

Feature	Semaglutide	Tirzepatide	Retatrutide
Mechanism	GLP-1 only	GLP-1 + GIP dual	GLP-1 + GIP + Glucagon triple
Weight Loss	~15-17%	~20-22%	-28.7% (68 wks)
>= 25% Loss	~10-15%	~36%	58.6%
Muscle Preservation	Moderate	Better (GIP)	Best (glucagon)
Visceral Fat	Significant	Superior	Most preferential
Metabolic Rate	No direct effect	Modest (GIP)	Significant increase
Liver Fat Reduction	~30-40%	~50-60%	>80%
Status	FDA approved	FDA approved	Phase 3 (2026)

LIVER FAT REDUCTION DATA

Sanyal et al., Nature Medicine 2024: MRI-PDFF analysis at 48 weeks: 8 mg retatrutide achieved -81.7% relative liver fat reduction; 12 mg achieved -86.0%. 89-93% of participants at highest doses reached liver fat <5% (MASLD resolution threshold). The glucagon receptor agonism drives hepatocyte CPT-1 alpha upregulation for enhanced beta-oxidation, simultaneously inhibiting SREBP-1c to reduce de novo lipogenesis.

IMPORTANT: Verify your retatrutide dose using a peptide reconstitution calculator before each injection. Titration should be overseen by a physician. This is not health advice.

STACKING NOTE:

Retatrutide creates the hormonal deficit signal (glucagon: mobilize fat; GLP-1: suppress appetite; GIP: enhance insulin sensitivity). When combined with 5-Amino 1MQ (NNMT inhibitor), the adipocyte mitochondria have sufficient NAD⁺ to execute the oxidation signaled by retatrutide. This creates a complementary demand-reduction + supply-side cellular capacity approach to fat loss. Non-overlapping mechanisms with the GH secretagogue stack.

ALTERNATIVES & ADD-ONS

CJC-1295 WITH DAC — LONG-ACTING GH SECRETAGOGUE

Dose	Frequency	Timing	Route
60 units	Once weekly	Any time (consistent day)	SubQ

MECHANISM: The Drug Affinity Complex (DAC) covalently bonds CJC-1295 to serum albumin via a maleimido-propionic acid linker, extending the half-life from ~30 minutes to 5.8-8.1 days. This creates sustained GH/IGF-1 elevation throughout the week, unlike the pulsatile pattern of No-DAC.

Feature	CJC-1295 WITH DAC	CJC-1295 NO DAC
Half-life	5.8-8.1 days	~30 minutes
GH Pattern	Sustained elevation (pulsatility preserved)	Acute pulsatile (mimics natural physiology)
Dosing	1x weekly	2x daily
Best For	Tissue repair, hypertrophy, recovery	Fat loss, sleep quality, long-term use
IGF-1 Elevation	1.5-3x above baseline for 9-11 days	Episodic; lower sustained level

WARNING: NEVER run both DAC and No-DAC simultaneously. No additive benefit — only additive receptor desensitization risk. Choose one version per cycle.

KEY STUDIES: Teichman et al., JCEM 2006 — Single dose raised GH 2-10x for 6+ days; IGF-1 1.5-3x for 9-11 days. Half-life 5.8-8.1 days. Ionescu & Frohman, JCEM 2006 (PMID: 17018654) — GH pulsatility preserved even with continuous GHRH stimulation (somatostatin oscillation maintained).

5-AMINO 1MQ — NNMT ENZYME INHIBITOR (NON-HORMONAL)

Dose	Frequency	Route	Cycle
50 mg	Daily	Oral capsule	4-8 weeks on / 2-4 weeks off

MECHANISM: Inhibits NNMT (nicotinamide N-methyltransferase) with an IC50 of ~1 uM. NNMT depletes NAD+ in adipocytes by diverting nicotinamide away from the salvage pathway. 5-Amino 1MQ restores intracellular NAD+ (1.2-1.6x increase), activating Sirt1 and AMPK — promoting mitochondrial biogenesis and fat oxidation at the adipocyte level. Non-hormonal: no testosterone, cortisol, thyroid, or estrogen modulation.

BENEFITS: Adipocyte-specific fat loss (5.1% weight loss in 11 days, mouse model), lean mass preservation, no appetite suppression (food intake identical between groups), NAD+ restoration, additive with exercise (~60% grip strength improvement when combined with training vs ~25% alone — Watowich et al., Scientific Reports 2024).

KEY STUDIES: Neelakantan et al., Biochem Pharmacol 2017 — Foundational study: concentration-dependent NAD+ increase; 5.1% weight loss in DIO mice; identical food intake; significant adipocyte size reduction. Watowich et al., Sci Rep 2024 — ~60% grip strength improvement with NNMT inhibition + exercise in sarcopenia model. No human RCTs as of March 2026.

MELANOTAN 1 + DAILY SCHEDULE

MELANOTAN 1 (AFAMELANOTIDE) — MC1R SELECTIVE AGONIST

Phase	Dose	Frequency	Duration
Loading	15 units	Daily	14 days
Maintenance	15 units	3x per week	Ongoing

MC1R SELECTIVITY: Unlike Melanotan 2 (which activates MC1R, MC3R, MC4R, MC5R causing sexual side effects, appetite suppression, and CNS effects), MT1 is highly selective for MC1R on melanocytes only. No CNS penetration, no sexual effects, no appetite changes. FDA/EMA approved as Scenesse for erythropoietic protoporphyria.

PHOTOPROTECTIVE MECHANISM: MC1R activation increases eumelanin production (brown/black, UV-protective) independent of UV exposure. Eumelanin caps form above cell nuclei, shielding DNA. Additionally, MC1R signaling enhances nucleotide excision repair (NER) pathways, suppresses NFkB inflammatory cascade, and increases catalase/ferritin antioxidant status. Pigmentation begins days 4-7; full photoprotection by day 14.

CLINICAL SAFETY: Post-authorization safety study (PASS), Germany 2016-2021 (N=200): 91% continued treatment; long-term TEAEs dropped to 11.3% (from 45.5% short-term). 8-year observational study (115 patients, 1,023 implants): good safety, no late effects. No melanoma link in clinical trials up to 8 years. Mole monitoring every 6-12 months recommended.

REMINDER: THIS IS NOT HEALTH ADVICE.

The schedule below is for educational reference only. Consult your physician before administering any compound.

Always use a fresh alcohol swab on every vial stopper and every injection site. Always verify doses with an online calculator.

DAILY INJECTION SCHEDULE OVERVIEW

Time	Compound	Dose	Notes
AM (Fasted)	CJC-1295 No DAC + Ipamorelin	20 units each	90 min fast after. Every day.
AM	GHK-Cu	10 units	Daily. Can combine with CJC/Ipa.
AM	5-Amino 1MQ (oral)	50 mg capsule	Daily oral. No injection required.
Pre-Workout	BPC-157	25 units	45 min before training. EOD only.
Pre-Workout	TB-500	25 units	EOD (offset from BPC-157 days).
PM (Before Bed)	CJC-1295 No DAC + Ipamorelin	20 units each	Fasted. Amplifies SWS GH pulse.
Weekly	Retatrutide	50-100 units	Same day each week. Ramp schedule.
Daily/3x wk	Melanotan 1	15 units	Loading daily x14, then 3x/wk.

EXPANDED SCHEDULE WITH LONGEVITY STACK

The schedule above covers the Core Stack only. After mastering the core compounds, layer in the Longevity Stack (NAD+/NMN, MOTS-c, Thymosin Alpha-1, Epithalon) using the full integrated schedule later in this guide. Core first. Always core first.

ALCOHOL SWAB COUNT PER SESSION (CORE STACK):

AM injection (CJC-1295 + Ipamorelin + GHK-Cu combined): Swab CJC vial (1) + Swab Ipa vial (2) + Swab GHK vial (3) + Swab injection site (4) = 4 swabs minimum

Pre-workout injection (BPC-157 OR TB-500): Swab vial (1) + Swab injection site (2) = 2 swabs minimum

PM injection (CJC-1295 + Ipamorelin): Swab CJC vial (1) + Swab Ipa vial (2) + Swab injection site (3) = 3 swabs minimum

TOTAL: 9-11 alcohol swabs per day on training days. Stock accordingly.

HEALTH & LONGEVITY PEPTIDES

Extending the Training Career // Cellular Health // Healthspan Optimization

You've built the foundation. Now you protect it. The compounds in this section target the biological machinery that determines **how long you train at a high level** — not just how hard you can push right now. Every natural athlete eventually confronts the ceiling: declining recovery, accumulating damage, immune vulnerability during hard blocks. This section addresses all of it.

These are not performance shortcuts. **They are longevity tools** — compounds targeting telomere biology, mitochondrial function, immune resilience, NAD+ metabolism, and sleep architecture. The goal is a decade more of productive training, not a badge on a forum.

WHY LONGEVITY PEPTIDES MATTER FOR NATURAL ATHLETES

◆ Training Career Extension

Every year of hard training depletes NAD+, shortens telomeres, and taxes immune function. These compounds slow that decline without exogenous hormones.

◆ Recovery Depth

MOTS-c and NAD+ precursors improve mitochondrial efficiency — the engine that drives repair between sessions. Poor mitochondrial function means incomplete recovery regardless of sleep quality.

◆ Immune Resilience

Caloric deficits and high training volumes suppress immune function. Thymosin Alpha-1 maintains T-cell and NK-cell activity during the most vulnerable periods — deep cuts and peak blocks.

◆ Cellular Quality Control

Epithalon (telomerase activation), SS-31 (cardiolipin stabilization), and Humanin (anti-apoptotic signaling) all target the cellular housekeeping that declines with age and training stress.

◆ Sleep Architecture

DSIP modulates slow-wave sleep depth — the same window where GH secretion peaks and tissue repair is maximized. Poor SWS quality is a ceiling on your entire recovery stack.

TIERED APPROACH — HOW TO READ THIS SECTION

These compounds are not equivalent in evidence quality. Read them in tier order:

TIER 1 — Strong Human Evidence: NAD+/NMN/NR (multiple RCTs), Thymosin Alpha-1 (FDA-approved dose, hepatitis B RCTs, sepsis meta-analysis)

TIER 2 — Mechanistic + Rodent/Limited Human: MOTS-c (one Phase 1 analog trial), Epithalon (cell culture + animal + 1 Soviet-era human study)

TIER 3 — Emerging/Educational: SS-31 (research setting IV only), Humanin (centenarian association), DSIP (sleep research context)

Deploy in that order. Do not skip to Tier 3 before Tier 1 is locked in.

CRITICAL DISCLAIMER: RESEARCH COMPOUNDS

MOTS-c, Epithalon, SS-31, Humanin, and DSIP are research compounds with no FDA approval for human use. NAD+ precursors (NMN/NR) are sold as dietary supplements — they are oral, not injectable. Thymosin Alpha-1 (Zadaxin) is FDA/EMA-approved at 1.6mg 2x/week — the dose referenced here.

THIS IS NOT HEALTH ADVICE. CONSULT YOUR PHYSICIAN BEFORE ANY USE.

THE INTEGRATION PRINCIPLE

Longevity compounds **amplify the core stack** — they do not replace it. The GH secretagogues build tissue. BPC-157 and TB-500 repair it. NAD+ and MOTS-c ensure the mitochondria have the fuel and signaling capacity to execute that repair at a cellular level. Epithalon and Thymosin Alpha-1 protect the cellular infrastructure that makes all of it possible. Run the core stack first. Add longevity compounds as a second layer.

COMPOUND OVERVIEW — AT A GLANCE

Compound	Route	Dose Range	Tier	Primary Mechanism
NAD+/NMN	Oral	500-1000mg daily	Tier 1	Sirtuin activation, mitochondrial biogenesis
MOTS-c	SubQ	5-10mg 3-5x/wk	Tier 2	AMPK activator, exercise mimetic
Epithalon	SubQ	5-10mg daily x10-20d	Tier 2	Telomerase activation, pineal regulation
Thymosin Alpha-1	SubQ	1.6mg 2x/wk	Tier 1	Immune modulation, T-cell maturation
SS-31	IV (research)	0.25mg/kg	Tier 3	Cardiolipin stabilization, mitochondrial
Humanin	Endogenous	N/A (research)	Tier 3	Neuroprotection, anti-apoptotic
DSIP	SubQ/IV	30-100mcg	Tier 3	Sleep optimization, cortisol modulation

MONTHLY COST CONTEXT:

- NMN (500mg/day, powder): \$30-60/month — most affordable entry point
- NR (Tru Niagen 300mg/day): \$40-50/month — best-studied NAD+ precursor brand
- MOTS-c (5mg 3x/week): \$80-150/month — research peptide pricing varies by vendor
- Epithalon (10mg/day x20 days, 2x/year): \$60-120 per cycle — very cost-effective per-year basis

- Thymosin Alpha-1 (1.6mg 2x/week): \$100-200/month — FDA-approved compound, higher quality control
- SS-31 / Humanin / DSIP: Research-grade only — pricing inconsistent, not recommended for routine budgeting

BLOODWORK ADDITIONS FOR LONGEVITY STACK:

- NAD+ metabolites (if available): Whole blood NAD+ levels — baseline and 8-week retest
- Telomere length (TruDiagnostic or similar): Baseline before Epithalon cycle, retest at 6 and 12 months
- Lymphocyte subsets (CD4/CD8 ratio): Before and during Thymosin Alpha-1 — tracks immune modulation
- hsCRP + IL-6: Inflammation markers — should decrease with NAD+/MOTS-c combination
- Fasting glucose + HOMA-IR: MOTS-c and NAD+ both affect insulin sensitivity — monitor for improvement

NAD+ OPTIMIZATION

NMN / NR — Oral Supplements // NOT Injectable // Mitochondrial Fuel

ROUTE OF ADMINISTRATION: ORAL CAPSULE OR POWDER — NOT INJECTABLE

NMN and NR are oral supplements. They are not peptides and are not injected. They are included here because they directly support the NAD+ demands of the peptide stack — particularly the MOTS-c/AMPK axis and the mitochondrial repair pathways activated by CJC/Ipa, BPC-157, and TB-500.

MECHANISM: NAD+ SALVAGE PATHWAY + SIRTUIN ACTIVATION

What NAD+ does: NAD+ is a cofactor for 500+ enzymatic reactions. It drives mitochondrial respiration (accepts electrons from glycolysis and the TCA cycle), activates sirtuins (SIRT1, SIRT3 — the longevity deacetylases), supports PARP-mediated DNA repair, and maintains the NAD+/NADH ratio essential for oxidative phosphorylation. NAD+ declines 10-65% across tissues with aging (Covarrubias et al., 2021).²⁷

Salvage pathway: NMN bypasses the NAMPT rate-limiting step — it is one enzymatic step from NAD+ (NMN → NAD+ via NMNAT). NR is two steps (NR → NMN → NAD+ via NRK1/2 then NMNAT). Both raise blood NAD+ in humans; NMN may have faster uptake via the Slc12a8 intestinal transporter.

Sirtuin chain: NMN/NR → ↑ NAD+ → SIRT1 activation → PGC-1α deacetylation → mitochondrial biogenesis (TFAM, NRF1, NRF2, COX subunits). This is the exact pathway MOTS-c (next page) also activates via AMPK — making them bidirectionally synergistic.

PARP competition: DNA damage activates PARP1, which outcompetes sirtuins for nuclear NAD+. High training loads + caloric restriction = chronically elevated PARP activation = NAD+ depletion. Oral NMN/NR restores the substrate for both PARP-mediated repair AND sirtuin-mediated longevity signaling simultaneously.

DOSING PROTOCOL (RESEARCH-DERIVED)

Compound	Dose Range	Timing	Route
NMN (β -NMN)	500–1,000 mg/day	Morning with food or fasted	Oral capsule/sublingual
NR (Nicotinamide Riboside)	300–1,000 mg/day	Morning (may split AM/PM)	Oral capsule
Choose ONE precursor	Start at lower end	Consistent daily timing	Not combined — pick NMN or NR

KEY HUMAN STUDIES

Yoshino et al., 2021 (Science):²⁰ Randomized, double-blind trial — 250 mg/day NMN in prediabetic women. Skeletal muscle insulin-stimulated glucose disposal increased ~25% (hyperinsulinemic-euglycemic clamp). AKT/mTOR phosphorylation improved in muscle biopsies. PDGF-receptor- β pathway and extracellular matrix remodeling genes upregulated via RNA-seq.

Yi et al., 2022 (GeroScience):²¹ Multicenter, double-blind RCT — 300, 600, or 900 mg NMN daily $\times$ 60 days (n=80). Blood NAD⁺ significantly elevated at all doses vs. placebo (all $p \leq 0.001$). Biological age (Aging.Ai 3.0) was preserved in NMN groups vs. a significant increase in placebo. 6-min walk distance improved at all doses.

Lapatto et al., 2023 (Science Advances):²² 5-month NR twin study — escalating to 1,000 mg/day. Skeletal muscle mitochondrial number increased ~14% by TEM; mtDNA copy number increased ~30%. SIRT1, TFAM, ERR α , MFN2, COX4 expression all increased. First human trial to confirm NR-driven mitochondrial biogenesis in muscle.

Liao et al., 2021 (J Int Soc Sports Nutr):²³ 6-week RCT in amateur runners — 300, 600, or 1,200 mg NMN daily + concurrent training. VO₂ at ventilatory threshold (VT1) and power at VT1/VT2 all significantly improved at 600-1,200 mg vs. placebo ($p < 0.05$). VO₂max unchanged — NMN improves submaximal muscular efficiency, not cardiac output ceiling.

SYNERGY: NAD⁺ + MOTS-c (AMPK AXIS)

NAD⁺ precursors and MOTS-c activate AMPK through different mechanisms — **bidirectional amplification**. NMN/NR raises NAD⁺ $\rightarrow$ SIRT1 $\rightarrow$ PGC-1 α $\rightarrow$ AMPK upstream signaling. MOTS-c activates AMPK directly via the Folate-AICAR pathway. Running both creates a convergent activation of the same longevity kinase from two independent routes. When combined with exercise (which also raises endogenous MOTS-c 11.9-fold in skeletal muscle), the three inputs are mechanistically additive.

ATHLETE-SPECIFIC NOTES

When NAD⁺ matters most for athletes: (1) During caloric restriction/cutting phases — NAD⁺ biosynthesis is reduced under caloric deficit. (2) During high training volume blocks — elevated PARP activation from DNA damage depletes NAD⁺. (3) Athletes 35+ years — NAMPT activity declines with age, slowing salvage pathway NAD⁺ regeneration. (4) During or after intense injury — PARP and CD38 hyperactivation both accelerate NAD⁺ depletion.

Important limitation (Campelj & Philp, 2022):²⁴ In healthy, well-trained young athletes, SIRT1 may already be sufficiently activated by exercise. The largest benefit is in athletes 35+, during caloric deficit, or during injury recovery — not necessarily in optimized, younger trainees in maintenance. **Do not expect a dramatic performance boost in peak training. Expect deeper recovery and longer**

career ceiling.

Safety (Song et al., 2023):²⁵ Across 10 published human NMN trials — doses up to 2,000 mg/day for 14 days — no serious adverse events. No changes in liver enzymes, renal function, hematology, ECG, or blood pressure. NR confirmed non-mutagenic (Ames test). Both are among the safest longevity supplements with human safety data.

NAD+ IV INFUSIONS — CURRENT EVIDENCE

NAD+ IV at 750 mg raised plasma NAD+ 398% at 6 hours but caused moderate-to-severe GI symptoms in all participants (Reyna et al., 2026). NR IV was better tolerated (37 min vs. 97 min infusion; minor symptoms only). Neither has robust RCT evidence. Commercial wellness IV protocols are outpacing the science. **Oral NMN/NR is the evidence-based route for daily use. IV is not recommended as routine practice.**

NAD+ PROTOCOL SUMMARY FOR ATHLETES

Recommended daily stack: NMN 500–1,000 mg/day (sublingual or enteric-coated capsule) + NR 300 mg/day (Tru Niagen or equivalent). Take in the morning on an empty stomach for optimal absorption. Some athletes add TMG (trimethylglycine) 500 mg to support methyl donor pools depleted by NAD+ metabolism.

Cycling recommendation: 8 weeks on / 2 weeks off. The off-cycle allows assessment of subjective recovery quality and prevents potential feedback inhibition of endogenous NAMPT activity. During the off-cycle, maintain resveratrol (150–300 mg) as an independent SIRT1 activator.

Bloodwork markers to track: Whole blood NAD+ levels (if available through specialty labs like Jinfiniti), fasting insulin, HbA1c, liver enzymes (AST/ALT), and inflammatory markers (hs-CRP, IL-6). NAD+ supplementation should show improvements in fasting insulin and inflammatory markers within 4–8 weeks in metabolically challenged individuals.

Storage: NMN is hygroscopic (absorbs moisture) and degrades at room temperature. Store sealed in a cool, dark place or refrigerate. NR is more shelf-stable but also benefits from refrigeration. Both degrade faster in heat and humidity — avoid leaving in gym bags or vehicles.

Cost estimate: Quality NMN (third-party tested, >99% purity): \$40–80/month at 500 mg/day. NR (Tru Niagen 300mg): ~\$40–50/month. Combined monthly cost: \$80–130. Budget-conscious athletes may choose NMN alone at 500 mg/day as the primary intervention.

MOTS-c

Mitochondrial-Derived Peptide // AMPK Activator // Exercise Mimetic

WHAT IS MOTS-c

MOTS-c (Mitochondrial Open reading frame of the 12S rRNA-c) is a 16-amino acid peptide encoded entirely within the mitochondrial genome — specifically the 12S rRNA gene. It was discovered by Lee et al. in 2015.³⁰ Unlike nuclear-encoded peptides, MOTS-c is translated directly from mitochondrial ribosomes and secreted into circulation, making it a true **mitokine** — a mitochondrial hormone that signals systemically.

MOTS-c is banned by WADA under S4 (Metabolic Modulators, section 4.4.1 — AMPK activators) because of its performance-enhancing, exercise-mimetic properties. This classification itself confirms its mechanistic potency.

MECHANISM: AMPK ACTIVATION + NUCLEAR TRANSLOCATION

Primary pathway: MOTS-c activates AMPK via the Folate → 10-formyl-THF → AICAR route. Specifically: MOTS-c inhibits the methylene-tetrahydrofolate dehydrogenase (MTHFD2) enzyme in the folate cycle, causing AICAR (ZMP) accumulation. AICAR is a natural AMPK activator. This is the same mechanism used by metformin (indirectly) and exercise.

Nuclear translocation (Kim et al., 2018):³¹ Under metabolic stress, MOTS-c translocates from the cytoplasm to the nucleus, where it binds ARE (antioxidant response elements) and NRF2 sites. This enables direct mitochondrial-to-nuclear communication — MOTS-c acts as a retrograde messenger that reprograms nuclear gene expression in response to mitochondrial status.

Downstream effects: AMPK activation → PGC-1α → mitochondrial biogenesis; SIRT1 → epigenetic regulation; NRF2 → antioxidant defense (HO-1, NQO1); HSF1 → heat shock protein expression and proteostasis. Simultaneously inhibits FOXO3a pro-apoptotic signaling.

DOSING: RESEARCH CONTEXT (NO APPROVED HUMAN PROTOCOL)

Dose	Frequency	Route	Cycle
5–10 mg	3–5x per week	SubQ injection	4–12 weeks on / 4–8 weeks off

CRITICAL: NO VALIDATED HUMAN DOSING EXISTS FOR NATIVE MOTS-c

Every practical dosing figure used in research communities is extrapolated from rodent intraperitoneal injections. Human-to-rodent dose translation is complex (body surface area conversion ~6:1) and has NOT been validated for MOTS-c. The only human safety data comes from CB4211 — a structurally distinct analog — from its Phase 1 trial, which reported injection site reactions (the most common TEAE) and limited systemic effects. **Treat as highly experimental.**

KEY STUDIES

Lee et al., 2015 (Cell Metabolism):³⁰ Discovery paper. MOTS-c promoted metabolic homeostasis, reduced obesity and insulin resistance in HFD mice without caloric restriction. Prevented diet-induced obesity; activated glucose metabolism in skeletal muscle; inhibited lipid biosynthesis in the liver. WADA banned it based on this data.

Kim et al., 2018 (Cell Metabolism):³¹ Nuclear translocation paper. MOTS-c physically translocates to the nucleus under metabolic stress, binds ARE promoter sites, and co-regulates adaptive nuclear gene expression. This established MOTS-c as a bidirectional mitochondrial-nuclear communicator, not merely a circulating hormone.

Reynolds et al., 2021 (Nature Communications):³² Human exercise data. A single moderate-intensity cycling bout raised skeletal muscle MOTS-c by **11.9-fold** in humans. MOTS-c levels decline with age. Late-life MOTS-c administration to aged mice (15 mg/kg, 3x/week) improved grip strength, treadmill endurance, and showed a trend toward increased lifespan. This is the most compelling human-relevant evidence for the exercise-MOTS-c axis.

BIDIRECTIONAL SYNERGY: MOTS-c + EXERCISE + NAD+

Direction	Mechanism	Practical Implication
Exercise → MOTS-c	Moderate-intensity aerobic exercise raises endogenous MOTS-c 11.9x (skeletal muscle)	Train consistently — you are already running the MOTS-c pathway
MOTS-c → Exercise Performance	MOTS-c enhances glucose metabolism, mitochondrial efficiency, reduces fatigue markers	Exogenous MOTS-c may amplify what exercise already initiates
NAD+ → MOTS-c signaling	Higher NAD+ supports SIRT1 → PGC-1α, which converges with MOTS-c/AMPK at the same targets	NMN/NR and MOTS-c are mechanistically additive on the AMPK/PGC-1α axis

SAFETY: CB4211 PHASE 1 CONTEXT

CB4211 is a MOTS-c analog (structurally modified, not native sequence). Its Phase 1 data showed: injection site reactions were the most common TEAE; no serious systemic adverse events; short-term safety confirmed. Self-reported effects with native MOTS-c from research communities include palpitations, insomnia, and fever — none confirmed in controlled settings. **The safety profile of native MOTS-c in humans is incompletely characterized.** Run shorter cycles (4 weeks), start at 5 mg 3x/week before progressing.

EPITHALON

Telomerase Activator // Pineal Gland Peptide // Longevity Research Compound

BACKGROUND

Epithalon (also spelled Epitalon; tetrapeptide Ala-Glu-Asp-Gly) was developed by the St. Petersburg Institute of Bioregulation and Gerontology, primarily under Vladimir Khavinson. It is a synthetic analog of Epithalamin — a polypeptide extract from the pineal gland. Its primary mechanism is **activation of telomerase (hTERT) and telomere elongation** — making it the most directly anti-aging peptide in this guide.

The evidence base is almost entirely from Soviet-era Russian research, cell culture, and animal models. **There are no large-scale, blinded human RCTs.** This context is non-negotiable: Epithalon belongs in Tier 2 (mechanistic + limited human evidence), not Tier 1. It is included here for educational completeness and because the mechanistic data is compelling enough that serious longevity researchers use it.

MECHANISM: hTERT ACTIVATION + PINEAL REGULATION

Telomerase pathway: Epithalon stimulates expression of hTERT — the catalytic subunit of telomerase — which adds TTAGGG repeats to telomere ends, preventing replicative senescence. Without telomerase activity, each cell division shortens telomeres by ~50-200 bp; sufficiently short telomeres trigger p53/p21 senescence pathways or apoptosis.

Pineal regulation: Epithalon appears to restore pineal gland melatonin secretion in aged subjects. In aged rats, it normalized the circadian disruption of melatonin and cortisol — both critical for sleep quality and recovery architecture. This creates a direct bridge between Epithalon and the sleep optimization goals that DSIP addresses.

DOSING PROTOCOL

Dose	Duration	Frequency	Cycles/Year
5–10 mg SubQ daily	10–20 consecutive days	Daily during cycle	2x per year (e.g., spring/fall)

Protocol rationale: Short intensive cycles (10-20 days) rather than chronic daily use. Telomere dynamics are slow — meaningful changes occur over months to years, not days. The short cycle approach is derived from Khavinson's animal protocols and empirical practice. Peptide is reconstituted in standard BAC water per the protocol on page 3.

KEY STUDIES

Khavinson et al., 2003 (Bull Exp Biol Med):³⁶ Epithalon induced telomerase activity and telomere elongation in human somatic cells (fibroblasts, embryonic kidney cells, fetal thymus cells) at concentrations of 0.1–100 nM. Cell division capacity increased. This is the foundational mechanistic paper. PMID: 12937682.

Al-dulaimi et al., 2025 (Biogerontology):³⁷ Epitalon increased telomere length in human cell lines — 4 to 26-fold increases in telomerase activity depending on cell type (HEK293, IMR-90, others). Mechanistically via hTERT upregulation. Also demonstrated activity through ALT (alternative lengthening of telomeres) pathways in cells without endogenous telomerase. PMC: PMC12411320.

Anisimov et al., 2003 (Neoplasma):³⁸ Female SHR mice — Epithalon treatment extended mean lifespan by +12.3% and maximum lifespan by +6.6% vs. controls. Reduced age-related changes in biomarkers. Slowed tumor incidence. PMID: 14501183. **This is animal data. Human lifespan extension is not established.**

Khavinson et al., 2000 (Mech Ageing Dev):³⁹ Epithalon extended lifespan in *Drosophila melanogaster* — a canonical aging research model — providing cross-species validation of the longevity signal. PMID: 11087911.

ATHLETE RELEVANCE + HONEST LIMITATIONS

Why athletes use it: Telomere length correlates with biological aging rate. Athletes who train hard for decades accumulate telomeric stress. Epithalon cycles 2x/year represent an attempt to slow this accumulation — extending the biological clock by targeting its most direct measurable mechanism.

Critical limitations: (1) No blinded human RCTs exist. (2) All human cell culture data is in vitro — not the same as in vivo pharmacokinetics in a healthy adult. (3) Most animal data comes from Khavinson's own group — limited independent replication. (4) Telomere length measurement is expensive and has high biological variability — self-monitoring response is impractical for most users.

Use with proportional caution.

EPITHALON PROTOCOL DETAILS

Standard protocol: 5–10 mg SubQ injection daily for 10–20 consecutive days. Cycle 1–2 times per year (typically January and July). This mirrors the original Khavinson research dosing schedules used in animal longevity studies.

Reconstitution: Reconstitute lyophilized Epithalon with bacteriostatic water (BAC water). Standard reconstitution: 2 mL BAC water into a 10 mg vial yields 5 mg/mL. Draw 1 mL (= 5 mg) or 2 mL (= 10 mg) per injection. Use a fresh alcohol swab on the vial stopper before every draw. Refrigerate reconstituted vials; use within 21 days.

Injection sites: Abdomen (2 inches from navel) or outer thigh. Rotate sites daily. Use 29–31 gauge insulin syringes. Pinch skin, insert at 45-degree angle, inject slowly, hold for 5 seconds before withdrawing. Apply gentle pressure with a clean alcohol swab after withdrawal.

What to monitor: There is no practical at-home test for telomere length. Instead, track surrogate biomarkers: sleep quality (deep sleep percentage on wearables), subjective recovery quality, inflammatory markers (hs-CRP), and immune function markers (lymphocyte subsets if available). The honest answer: you will not "feel" Epithalon working in the short term. This is a long-term investment in biological aging rate.

Cost estimate: Epithalon 10 mg vials: \$25–50 each from reputable peptide sources. A 20-day cycle at 10 mg/day = 20 vials = \$500–1,000 per cycle. Two cycles per year: \$1,000–2,000 annually. This is among the more expensive peptides in the protocol but requires the least frequent dosing (only 20–40 injection days per year total).

THYMOSIN ALPHA-1 (Tα1)

Immune Sentinel // T-Cell Maturation // FDA-Approved Dose Available (Zadaxin)

WHAT IS THYMOSIN ALPHA-1

Thymosin Alpha-1 (Tα1) is a 28-amino acid peptide naturally produced by the thymus gland, where it orchestrates T-cell maturation. It is FDA/EMA-approved as Zadaxin (1.6 mg SubQ, 2x/week) for hepatitis B and adjuvant cancer immunotherapy in multiple countries. Of all compounds in the longevity section, **Tα1 has the strongest and most clinically validated evidence base** — multiple RCTs, sepsis meta-analyses, and active use in clinical oncology.

The thymus involutes (shrinks and loses function) beginning in the mid-20s, accelerating after age 40 — progressively depleting the output of new naive T-cells. **Immunosenescence** — the aging of the immune system — is one of the primary mechanisms of age-related disease vulnerability. Tα1 partially compensates for thymic involution, maintaining immune surveillance capacity independent of thymic output.

MECHANISM: T-CELL MATURATION + TLR AGONISM

T-cell maturation: Tα1 promotes differentiation of immature thymocytes into functional CD4+ T-helper and CD8+ cytotoxic T-cells. It upregulates T-cell surface markers (CD25, CD71), increases IL-2 receptor expression, and potentiates antigen-specific T-cell responses.

NK cell activation: Tα1 increases natural killer (NK) cell cytotoxicity and NK-cell numbers. NK cells are the first-line surveillance against viral infection and cancer cell recognition.

TLR-2 and TLR-9 agonism: Tα1 directly activates Toll-Like Receptors 2 and 9 on dendritic cells, triggering innate immune priming. TLR-9 agonism is particularly important for anti-viral immunity and the cross-priming of adaptive T-cell responses (Mathew et al., 2020).⁴¹

Immunomodulation, not immunosuppression: Critical distinction — Tα1 does not suppress immune function (like corticosteroids). It restores normal immune activity where it has been depressed by caloric deficit, overtraining, or aging. It does not drive autoimmune risk.

DOSING — ZADAXIN REFERENCE DOSE

Dose	Frequency	Route	Duration
1.6 mg	2x per week	SubQ injection	12–52 weeks (disease context); 8–12 weeks (training context)

Standard vial reconstitution: Zadaxin 1.6 mg vials come pre-configured for clinical use. Research-grade Tα1 follows the same BAC water reconstitution standard (page 3). At 1.6 mg per injection, 2x/week = 3.2 mg/week — well within the safety data from clinical trials.

KEY EVIDENCE

Hepatitis B RCTs: Multiple randomized controlled trials established 1.6 mg SubQ 2x/week as the effective dose for chronic hepatitis B — the basis for FDA approval as Zadaxin. Response rates (HBe-Ag seroconversion, viral suppression) consistently superior to placebo in chronic hepatitis B.

Sepsis meta-analysis (Liu et al., 2016, BMC Infect Dis):⁴² Systematic review of Tα1 RCTs in sepsis — RR 0.59 for 28-day mortality (41% relative risk reduction vs. standard care). Tα1-treated patients showed faster immune function recovery (lymphocyte counts, HLA-DR expression on monocytes). This is the most dramatic mortality data for any peptide in this guide.

COVID-19 (Diao et al., 2020, Clin Infect Dis):⁴³ Tα1 reduced 28-day mortality and reversed lymphocytopenia in severe COVID-19 (n=76 severe patients). CD4+, CD8+ T-cells and NK cells recovered faster in the Tα1 arm. Exhausted T-cell markers (PD-1, TIM-3) were reduced. PMC: PMC7314217.

Aging/immunosenescence (Simonova et al., 2025, IJMS):⁴⁴ Comprehensive review — Tα1 upregulates thymopoiesis markers, restores naive T-cell output, and improves vaccine responsiveness in elderly populations. Supports use as a preventive immunological intervention in aging athletes.

PRACTICAL ATHLETE APPLICATION

Scenario	Rationale	Priority
Deep caloric deficit (cut phase)	Caloric restriction suppresses T-cell proliferation and NK activity; Tα1 compensates	HIGH
High-volume training blocks	Overreaching suppresses immune function for 24-72h post-session ('open window'); Tα1 reduces this window	HIGH
Travel / competition exposure	Immune challenge during travel; Tα1 maintains NK and T-cell readiness	MEDIUM
Aging athletes (35+)	Progressive immunosenescence; Tα1 partially replaces declining thymic output	HIGH
Post-illness recovery	Reestablish baseline immune surveillance faster; historically used for chronic infections	MEDIUM

THYMOSIN ALPHA-1 — BOTTOM LINE

The strongest evidence in this longevity section. Approved dose. Multiple RCTs. Sepsis mortality data. Best deployed during caloric deficits, high training volume, and in athletes 35+. Not for acute performance enhancement — for maintaining the immune infrastructure that keeps you training consistently.

SS-31 // HUMANIN // DSIP

Emerging Mitochondrial Peptides + Sleep Optimization // Educational Reference

TIER 3 — EMERGING / LIMITED HUMAN DOSING DATA

SS-31 dosing in published human trials uses IV administration in research settings — not self-administered SubQ. Humanin has no established self-administration dosing protocol. DSIP has oldest research base but inconsistent human data. This page is **educational reference only** — not a self-administration guide. All require physician oversight.

SS-31 (ELAMIPRETIDE) — CARDIOLIPIN STABILIZER

What it is: SS-31 (H-D-Arg-Dmt-Lys-Phe-NH₂) is a Szeto-Schiller peptide that selectively partitions to the inner mitochondrial membrane and binds cardiolipin — a phospholipid essential for cristae structure, Complex I–IV assembly, and the cytochrome c-electron transfer chain. It is being developed as Elamipretide (Stealth BioTherapeutics) for primary mitochondrial myopathy and Barth syndrome.

Mechanism (Birk & Szeto, 2014;⁴⁶ Mitchell et al., 2020⁴⁷): SS-31 stabilizes the cardiolipin-cytochrome c interaction at the inner membrane, preventing electron leak (superoxide generation) while maintaining electron transport chain efficiency. Cardiolipin oxidation by cytochrome c — a pro-apoptotic signal — is blocked. Net effect: reduced ROS, improved ATP generation efficiency, mitochondrial morphology normalization.

Human evidence (Roshanravan et al., 2021, JCI Insight⁴⁸ — NCT02245620): Randomized trial in older adults — single IV dose of SS-31 (0.25 mg/kg) increased in vivo skeletal muscle mitochondrial ATP production (measured by 31P-MRS) in older adults. The effect was statistically significant after a single dose. This is the most compelling human SS-31 data for muscle relevance.

Context: The 0.25 mg/kg IV dose in a research setting is not equivalent to self-administered SubQ use. **No validated SubQ dosing protocol exists.** Elamipretide Phase 2 trials use IV infusion. Pending FDA review for primary mitochondrial myopathy.

HUMANIN — NEUROPROTECTIVE MITOCHONDRIAL-DERIVED PEPTIDE

What it is: Humanin (HN) is a 21-amino acid peptide encoded within the 16S rRNA gene of the mitochondrial genome — making it a mitochondrial-derived peptide (MDP), similar to MOTS-c but with distinct targets. It was discovered as a neuroprotective factor against Alzheimer's disease amyloid beta toxicity.

Mechanism: Humanin activates two receptor systems: (1) CNTFR/gp130/WSX-1 trimer → JAK2/STAT3 → cytoprotective gene expression; (2) FPRL1/FPRL2 GPCRs → ERK1/2 → anti-apoptotic signaling. It inhibits Bax-mediated apoptosis, suppresses inflammatory cytokines (IL-6, TNF-α), and protects neurons from amyloid beta, oxidative stress, and ischemia.

Centenarian association (Yen et al., 2020, Aging⁵⁰): Circulating Humanin levels are significantly higher in centenarian offspring vs. age-matched controls — independent of multiple confounders. Humanin levels decline with normal aging but are preserved in long-lived individuals. This association is one of the strongest epidemiological arguments for its longevity relevance in humans.

Limitation: No established self-administration dosing protocol. Research dose ranges come from rodent models only. Including Humanin here because the centenarian data and mechanism are scientifically significant — not because a practical SubQ protocol can be responsibly recommended.

DSIP — DELTA SLEEP-INDUCING PEPTIDE

What it is: DSIP is a nonapeptide (9 amino acids) originally isolated from rabbit cerebral venous blood perfusate in 1974 during studies on sleep-inducing factors. It crosses the blood-brain barrier (Liu et al., 2024⁵³) and acts on sleep architecture by modulating delta wave (slow-wave, SWS) activity — the same sleep phase during which GH pulsatility peaks.

Mechanism: DSIP modulates cortisol secretion (reducing stress-induced hypercortisolism — Sudakov et al., 1983⁵⁴), protects against mitochondrial electron transport chain dysfunction under hypoxia (Khvatova et al., 2003⁵⁶), and in clinical pilot data (Larbig et al., 1984⁵⁵) reduced pain episodes in chronic pain patients with concurrent sleep improvement.

Athlete relevance: GH pulsatility during SWS is the primary biological mechanism the CJC/Ipa PM dose amplifies. If DSIP deepens SWS, it amplifies the recovery value of the PM injection. Cortisol reduction supports the anabolic signaling environment. The mechanism is plausible; the clinical data is old, small, and methodologically limited.

Limitations: No modern large-scale RCTs. Most DSIP research is 1970s-1990s. Dosing varies widely in practice (100-500 mcg SubQ) with no validated human protocol from peer-reviewed studies. Short half-life (~4-5 min in blood, though effects persist). Use is primarily empirical in longevity research communities.

EMERGING COMPOUNDS — EVIDENCE SUMMARY

Compound	Best Human Evidence	Research Gap	Status
SS-31	Single-dose RCT: ↑ skeletal muscle ATP production (Roshanravan 2021)	No SubQ dosing; IV only; Phase 2 trials ongoing	Research setting only
Humanin	Centenarian epidemiology: higher levels → longer healthspan (Yen 2020)	No human dosing RCT; no SubQ protocol	Educational only
DSIP	Pilot: SWS improvement, cortisol modulation, pain reduction (Larbig 1984)	Old data; no modern RCTs; no validated dose	Empirical use only

THE TIER 3 PRINCIPLE

These compounds represent the scientific frontier — compelling mechanisms, emerging evidence, and significant biological plausibility. They are not ready for routine self-administration protocols. The right approach: lock in your core stack + NAD+ + MOTS-c + Tα1 first. These compounds become relevant only after years of protocol consistency and physician oversight. Know them. Follow the research. Do not rush them.

EXPANDED DAILY SCHEDULE

Core Stack + Longevity Stack + Oral Support // Full Protocol Integration

REMINDER: THIS IS NOT HEALTH ADVICE.

This schedule is for research and educational reference only. Consult your physician before administering any compound. Always use a fresh alcohol swab on every vial stopper and injection site. Verify all doses with an online reconstitution calculator.

TIER 1 — CORE STACK (Pages 4-9)

Time	Compound	Dose	Route	Notes
AM (Fasted)	CJC-1295 No DAC + Ipamorelin	20u each	SubQ	90 min fast after. Every day.
AM	GHK-Cu	10u	SubQ	Daily. Can combine with CJC/Ipa syringe window.
AM	5-Amino 1MQ (oral)	50 mg	Oral	Daily oral capsule. No injection.
Pre-Workout	BPC-157	25u	SubQ	45 min pre-training. EOD only.
Pre-Workout	TB-500	25u	SubQ	EOD (offset BPC-157 days preferred).
PM (Bed)	CJC-1295 No DAC + Ipamorelin	20u each	SubQ	Fasted. Amplifies SWS GH pulse.
Weekly	Retatrutide	50-100u	SubQ	Same day each week. Titration ramp.
Daily/3x wk	Melanotan 1	15u	SubQ	Loading daily x14, then 3x/wk.

TIER 2 — LONGEVITY STACK (Pages 10-14)

Time	Compound	Dose	Route	Notes
AM (with food)	NMN or NR (oral)	500-1,000 mg	Oral	Daily. Choose NMN or NR — not both.
AM (fasted SubQ)	MOTS-c	5-10 mg	SubQ	3-5x/week. 4-12 week cycles. Highly experimental.
Cycle 2x/year	Epithalon	5-10 mg/day	SubQ	10-20 consecutive days per cycle. Spring/fall.
2x per week	Thymosin Alpha-1	1.6 mg	SubQ	Standard Zadaxin dose. Best during cut or high volume.

TIER 3 — ORAL SUPPORT (Non-Peptide, Daily)

Compound	Dose	Timing	Purpose
TUDCA	500 mg	AM with food	Liver protection (essential with any stack)
NAC	1,200 mg	AM/PM split	Glutathione precursor; liver + antioxidant support
Omega-3	3-4 g EPA/DHA	With largest meal	Anti-inflammatory, lipid optimization
Vitamin D3 + K2	5,000 IU D3 / 200 mcg K2	With fat-containing meal	Hormonal support, calcium metabolism
Magnesium Glycinate	400-600 mg	PM before bed	Sleep quality, muscle relaxation, SWS support

CoQ10 (Ubiquinol)	200-400 mg	With fat-containing meal	Mitochondrial electron transport chain cofactor
Creatine Monohydrate	5 g	Post-workout or AM	Performance, cellular hydration, cognitive support

FULL STACK INTEGRATION PRINCIPLES

Do not start everything at once. Core stack first — stabilize 8-12 weeks. Then add Tα1 (especially during a cut). Then add NMN/NR (daily, low commitment). Add MOTS-c after you have bloodwork baseline and understand your individual response to the core stack. Epithalon cycles 2x/year regardless of what else you are running.

The longevity compounds do not replace the core. CJC/Ipa, BPC-157, TB-500, GHK-Cu, and retatrutide remain the functional performance stack. The longevity compounds extend the environment in which that stack operates — mitochondrial fuel (NAD+), exercise-mimetic cellular signaling (MOTS-c), immune maintenance (Tα1), and cellular renewal (Epithalon).

Bloodwork priority escalates with stack complexity. Running the core stack requires bloodwork every 90 days. Adding MOTS-c, NMN, and Tα1 does not change this schedule but adds IGF-1, fasting insulin, immune markers (CBC with differential, lymphocyte subsets), and optionally NAD+ metabolites (NAAD, NMN in blood) if available at your lab.

SYNERGY MAP — LONGEVITY COMPOUNDS + CORE STACK

Longevity Compound	Synergy With	Combined Mechanism
NMN/NR	MOTS-c	Both activate AMPK via convergent pathways — NAD+/SIRT1 and AICAR/AMPK
NMN/NR	CJC/Ipa + BPC-157	NAD+ fuels mitochondrial repair demanded by GH-driven anabolism
MOTS-c	Exercise	Exercise raises MOTS-c 11.9-fold; exogenous MOTS-c amplifies and extends that signal
Thymosin Alpha-1	Retatrutide (cut)	Reta drives caloric deficit; Tα1 compensates for cut-induced immune suppression
Epithalon	CJC/Ipa (PM dose)	Epithalon restores pineal melatonin rhythm; CJC/Ipa amplifies the SWS GH pulse within that restored architecture
All Longevity	All Core	Core stack builds. Longevity stack protects the cellular infrastructure that makes building possible.

BLOODWORK INTERPRETATION GUIDE

Protocol-Specific Reference Ranges & Red Flags

CRITICAL CAVEATS: High protein (200g+/day) artificially elevates BUN/creatinine — use Cystatin C for true kidney assessment. High muscle mass + training elevates AST independent of liver — use GGT as the differentiator. GH secretagogues increase fasting insulin; monitor HOMA-IR as the composite metric.

Marker	Normal Range	Protocol Effect	Red Flag	Action
ALT	7-56 U/L	NAC/TUDCA lower; training may raise	>3x ULN (>168)	Pause supps; rule out hepatitis
AST	10-40 U/L	Training raises (muscle); NAC lowers	>3x ULN + GGT up	Check CK to differentiate muscle vs liver
GGT	9-48 U/L (men)	TUDCA/NAC lower; KEY differentiator	>2x ULN (>96)	True liver signal — stop hepatotoxic agents
BUN	7-25 mg/dL	High protein raises (25-40 expected)	>50 + Cystatin C up	Only act if paired with Cystatin C elevation
Creatinine	0.74-1.35	Muscle mass + protein elevate (1.2-1.6 normal)	>2.0 + Cystatin C up	Defer draw 48h post-training; confirm w/ Cystatin C
eGFR	>90 optimal	Protein/muscle lower by 10-20 pts	<60 (Cystatin C)	Nephrology consult if Cystatin C-based eGFR <60
Fasting Glucose	70-99 mg/dL	GH peptides raise; reta lowers (-23.5)	>126 x2 draws	Reduce CJC/Ipa frequency; verify reta dose
HbA1c	<5.7%	GH: neutral at low dose; reta: -0.91%	>=6.5%	Full diabetes eval; pause GH peptides
Total T	300-1100 ng/dL	Fadogia/Tongkat raise endogenously	>1200 sustained	Cycle off Fadogia; reduce Tongkat Ali
Free T	4.85-19 ng/dL	Tongkat Ali liberates from SHBG	>25 ng/dL	Assess E2 aromatization; reduce dose
SHBG	10-57 nmol/L	Tongkat/GH axis lower SHBG	<10 nmol/L	Excess free androgen — check liver
IGF-1	82-244 ng/mL	CJC/Ipa raise +50-150 ng/mL	>450 ng/mL	Reduce CJC/Ipa to once daily; target 250-350
LDL	<100 optimal	Cut + omega-3 + reta lower	>160 mg/dL	Dietary fat quality review
HDL	>40 (men)	CoQ10 raises; severe cut may lower	<35 mg/dL	Increase healthy fats; optimize CoQ10
Triglycerides	<150 mg/dL	Omega-3/reta/cut all lower	>500 mg/dL	Pancreatitis risk — increase omega-3
hsCRP	<1.0 mg/L	GHK-Cu/BPC-157/NAC/omega-3 lower	>10 mg/L	Rule out infection; audit stack

TESTING SCHEDULE & ESTIMATED COSTS

Timepoint	Priority Tests	Est. Cost
Baseline (Pre-Protocol)	Full panel: CMP, CBC, lipids, ApoB, hormones, IGF-1, fasting insulin, HbA1c, hsCRP, Vit D, Cystatin C, GGT	\$680-960
Week 4-6 (Early Check)	CMP, GGT, Cystatin C, IGF-1, fasting insulin/glucose, testosterone/SHBG/E2	\$350-550
Post-Cycle (Week 13+)	Full baseline repeat + HOMA-IR calc + thyroid + AM cortisol	\$680-960

RED FLAGS & NON-NEGOTIABLES

STOP & ASSESS IMMEDIATELY IF YOU EXPERIENCE

■ Persistent nausea/vomiting

Beyond initial retatrutide titration period (2-3 weeks). If severe or worsening, pause and consult physician.

■ Dark urine or clay-colored stool

Potential liver/biliary signal. Check liver panel immediately (ALT, AST, GGT, bilirubin).

■ Jaundice (yellowing of skin/eyes)

Urgent hepatic evaluation. Stop all supplements immediately. Maintain NAC + TUDCA only.

■ Extreme fatigue / unusual lethargy

Beyond normal training fatigue. May indicate hormonal disruption, hepatic stress, or hypoglycemia.

■ Heart palpitations / irregular heartbeat

May indicate electrolyte imbalance (low magnesium/potassium) or cardiac concern. Seek evaluation.

■ Changing moles (ABCDE criteria)

New moles, changing shape/color, asymmetry, irregular borders. Especially critical on MT1. Dermatologist ASAP.

■ Hypoglycemia symptoms

Shakiness, confusion, sweating when fasted. Monitor glucose during GH peptide + retatrutide combination.

■ Injection site infection

Redness, warmth, swelling, pus at injection site persisting >48 hours. Seek medical evaluation.

THE NON-NEGOTIABLES

✓ 200g+ PROTEIN DAILY

Non-negotiable foundation. Lean sources. Spread across 4-5 meals. Supports muscle preservation during cut and peptide-enhanced recovery.

✓ 7.5-8 HOURS SLEEP

GH is primarily released during slow-wave sleep. The PM CJC/lpa dose amplifies the largest GH pulse of the day. Poor sleep = wasted peptides.

✓ 1 GALLON WATER DAILY

Hydration supports kidney filtration (critical with high protein), peptide absorption, and metabolic processes.

✓ LIVER SUPPORT (TUDCA + NAC)

TUDCA 500mg + NAC 1200mg daily. Active hepatoprotection. These are not optional — they actively lower liver stress markers.

✓ CARBS AROUND TRAINING

Glycogen fuels performance. Time carbohydrate intake around the training window for energy and recovery.

✓ BLOODWORK: MANDATORY

Baseline before starting. Week 4-6 check. Post-cycle full panel. You cannot manage what you don't measure.

THE BOTTOM LINE

If something feels wrong, it probably is. Trust the data over your feelings — bloodwork does not lie.

No peptide is worth compromising your health. Pause the protocol, get labs drawn, and consult your physician before resuming. You can always restart. You cannot undo organ damage.

"Peptides are the accelerator. You are the engine."

FINAL REMINDER: THIS IS NOT HEALTH ADVICE.

Everything in this document is for research and educational purposes only. The author is not a medical professional. You are solely responsible for your own health decisions. Before starting, adjusting, or stopping any protocol: **CONSULT YOUR DOCTOR.** No exceptions.

QUICK-REFERENCE: WHEN TO STOP A PEPTIDE

Stop immediately and consult your physician if you experience any of the following:

- Persistent injection site reactions (redness, swelling, warmth lasting >48 hours) — possible infection or allergic response
- Unexplained fatigue, nausea, or malaise not attributable to training load or diet changes
- Dark urine, jaundice (yellowing of skin/eyes), or upper-right abdominal pain — potential liver stress
- Heart palpitations, chest tightness, or irregular heartbeat — cardiac evaluation required
- Sudden water retention, facial puffiness, or rapid weight gain (>5 lbs in 1 week) — possible hormonal dysregulation
- Elevated liver enzymes (AST/ALT >2x upper limit), elevated fasting glucose, or abnormal kidney markers on bloodwork
- Numbness, tingling, or carpal tunnel symptoms (particularly with GH-elevating peptides) — reduce dose or discontinue
- Any symptom you cannot explain. When in doubt, stop the protocol and get labs drawn. You can always restart. You cannot undo organ damage.

PEPTIDE SCIENCE APPENDIX

Interaction Map + Vial Calculator + Stack Synergy

PEPTIDE INTERACTION MAP

CJC-1295 + IPAMORELIN

GHRHR + GHSR-1a -> Synergistic GH Pulse -> IGF-1 elevation

GHK-Cu

~4,000 gene modulation (~30% genome)
TGF-beta, NFkB suppression, DNA repair

BPC-157

Src-eNOS (NO), FAK-paxillin, JAK2/GHR
Vascular scaffolding + GH receptor upreg

TB-500

G-actin sequestration, ILK/Akt pathway
Cell migration + epicardial progenitor activation

ALL 4: Sequential, non-redundant pathway engagement across full healing cascade

VIAL DURATION CALCULATOR (5MG VIAL, 300 UNITS BAC WATER)

Peptide	Dose	Frequency	Units/Week	Vial Duration
CJC-1295 No DAC	20 units	2x daily (14x/wk)	280 units	~1.1 weeks/vial
Ipamorelin	20 units	2x daily (14x/wk)	280 units	~1.1 weeks/vial
GHK-Cu	10 units	Daily (7x/wk)	70 units	~4.3 weeks/vial
BPC-157	25 units	EOD (~3.5x/wk)	87.5 units	~3.4 weeks/vial
TB-500	25 units	EOD (~3.5x/wk)	87.5 units	~3.4 weeks/vial
Retatrutide	50-100 units	1x weekly	50-100 units	3-6 weeks/vial
Melanotan 1	15 units	Daily/3x wk	105/45 units	2.9/6.7 wks

DOSING CALCULATOR — VERIFY EVERY DOSE BEFORE YOU INJECT

CRITICAL: ALWAYS VERIFY YOUR DOSING MATH.

The vial duration table above assumes a standard 5mg vial reconstituted with 300 units (3mL) BAC water. Your vials may differ in peptide amount or reconstitution volume. **Never guess your dose.** Use an online peptide reconstitution calculator to verify your exact draw volume based on YOUR specific setup.

THIS IS NOT HEALTH ADVICE. CONSULT YOUR PHYSICIAN TO CONFIRM ALL DOSING.

RECOMMENDED ONLINE CALCULATORS: Use one of these free tools to cross-check your dosing math before every new vial or any time you change your reconstitution volume. Enter your vial strength (mg), BAC water volume (mL), and desired dose (mcg) — the calculator tells you exactly how many units to draw on your insulin syringe.

✓ **Omni Calculator — Peptide Dosage** — omnicalculator.com/health/peptide-dosage — Clean, ad-free, shows syringe pull units + total cycle supply calculation. Supports body-weight-based dosing.

✓ **PepCalc App (iOS/Android)** — Available on App Store and Google Play. Professional-grade reconstitution and dosage calculator with insulin syringe unit conversion. Free, no ads.

✓ **Cellgenic Peptide Calculator** — cellgenic.com/peptide-calculator — Visual syringe dosage meter. Select dose, vial strength, and water volume for instant results.

✓ **AI-Assisted Verification** — Use a large language model (e.g., Claude Opus 4.6 or similar) to double-check your reconstitution math. Describe your vial concentration, BAC water volume, and desired mcg dose — ask it to calculate the exact syringe draw in units. Cross-reference with the calculator above.

HOW TO USE: (1) Enter your peptide vial amount (e.g., 5mg). (2) Enter the volume of BAC water you added (e.g., 3mL / 300 units). (3) Enter your desired dose in mcg. (4) The calculator returns the exact number of units to draw on a U-100 insulin syringe. **Always confirm with your physician.**

STACK SYNERGY SUMMARY

Combination	Synergy Mechanism	Outcome
CJC/Ipa + BPC-157	BPC-157 upregulates GH receptor; IGF-1 potentiates JAK2/GHR	Amplified connective tissue repair
CJC/Ipa + GHK-Cu	GH stimulates collagen + GHK-Cu drives TGF-beta independently	Additive matrix remodeling
BPC-157 + TB-500	BPC: signaling (NO/VEGF); TB: cell migration into environment	"Gold standard" repair
GHK-Cu + BPC-157	Both suppress NFkB/IL-6 via different pathways	Additive anti-inflammatory
Reta + 5-Amino 1MQ	Reta: hormonal deficit; 5A1MQ: adipocyte NAD+ restoration	Demand + supply-side fat loss
All 4 Core	Sequential, non-redundant pathway engagement	Comprehensive tissue optimization

OPTIMIZATION PRINCIPLES

1. Verify Before You Inject: Always cross-check your dosing math with an online peptide reconstitution calculator (see above). When in doubt, use a second calculator or AI tool to confirm. A dosing error with a weekly injectable like retatrutide can mean a week of incorrect exposure.

2. One Variable at a Time: When adding a new compound to your stack, introduce it alone for 2-4 weeks before adding the next. This lets you isolate any side effects or benefits to a specific compound. If you start everything at once and experience an issue, you will not know which compound caused it.

3. Bloodwork Is Your Dashboard: You are running an experiment on yourself. The only way to know if it is working — and if it is safe — is to measure. Baseline labs before starting, check at week 4-6, full panel post-cycle. This is not optional.

4. Document Everything: Keep a daily log of what you inject, how much, which site, and any effects (positive or negative). This data is invaluable when consulting your physician and when optimizing your protocol over time.

THIS IS NOT HEALTH ADVICE.

All dosing, timing, and stacking information is for research and educational purposes only. You **MUST** consult a licensed physician before beginning, adjusting, or combining any compounds.



*"Seventeen years under the bar
before I ever picked up a syringe.*

*Earn the foundation first.
Then build on it."*

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